

Group Representations

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Preface

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Chapter 1

Introduction

Group representation theory studies groups by letting them act on vector spaces via linear maps, thereby *representing* abstract group elements as concrete invertible matrices. This allows the full power of linear algebra to be brought to bear on group-theoretic problems. The present chapter introduces the two key notions: group actions on sets, and linear representations as group actions on vector spaces.

1.1 Group Actions

The *Erlangen programme*, introduced by Felix Klein in 1872, proposes that groups do not exist in the void—they are meant to move the points of *spaces*. We obtain a much clearer understanding of a group by specifying what it acts on.

Example 1.1. • The linear group $\mathrm{GL}_n(\mathbb{R})$ ($n \in \mathbb{N}$) moves the points of the vector space $\mathbb{R}^n$ by matrix multiplication.

- The symmetric group S_n ($n \in \mathbb{N}$) permutes the elements of the set $\{1, 2, \dots, n\}$.
- The cyclic group $C_n = \mathbb{Z}/n\mathbb{Z}$ ($n \in \mathbb{N}$) can be thought of as the group of rotations of the plane which stabilise a regular n -gon, acting on its vertices.

Definition 1.2 (Group action). Let G be a group and X be a set. An *action* $G \curvearrowright X$ of G on X is a map

$$G \times X \longrightarrow X, \quad (g, x) \longmapsto g \cdot x,$$

satisfying

- $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$,
- $1_G \cdot x = x$ for all $x \in X$.

Example 1.3. A Rubik's cube is not a group, but a set X of configurations on which a group of motions G acts. The group structure lives in G , not in X .

Remark 1.4. The axioms imply that for each $g \in G$ the map

$$\rho_g: X \longrightarrow X, \quad x \longmapsto g \cdot x$$

is a bijection with inverse $\rho_{g^{-1}}$, since $\rho_{g^{-1}} \circ \rho_g = \rho_{1_G} = \text{id}_X$. Thus an action of G on X is equivalently a group morphism $G \rightarrow \text{Sym}(X)$, where $\text{Sym}(X)$ denotes the group of all bijections $X \rightarrow X$.

1.2 Linear Representations

A linear representation is a group action on a set that carries the additional structure of a vector space, with the requirement that every group element acts by a linear map.

Definition 1.5 (Linear representation). Let K be a field and $n \in \mathbb{N}$. A (linear) representation of G over K of degree n is a group morphism

$$\rho: G \longrightarrow \text{GL}_n(K).$$

So a representation of G is an action of G on a set which is a vector space, such that every $g \in G$ acts by a linear transformation. The elements of G are thereby *represented* by invertible matrices.

Example 1.6. • $\text{GL}_n(\mathbb{R}) \curvearrowright \mathbb{R}^n$ is actually a representation of $\text{GL}_n(\mathbb{R})$: the identity map $\text{GL}_n(\mathbb{R}) \rightarrow \text{GL}_n(\mathbb{R})$ is a group morphism of degree n over $\mathbb{R}$.

- The map

$$\mathbb{C}^\times \longrightarrow \text{GL}_2(\mathbb{R}), \quad a + bi \longmapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is a representation of $\mathbb{C}^\times$ of degree 2 over $\mathbb{R}$.

- For any field K , the map

$$(K, +) \longrightarrow \text{GL}_2(K), \quad x \longmapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

is a representation of the additive group $(K, +)$ of degree 2 over K .

- The *trivial representation*: the constant map $G \rightarrow \text{GL}_n(K)$, $g \mapsto I_n$, is a representation of degree n for any G , K , and n .

Coordinate-free formulation

It is often convenient to work with a vector space V without fixing a basis.

Definition 1.7. Alternatively, a representation of G over K is a group morphism

$$\rho: G \longrightarrow \text{GL}(V),$$

where V is a K -vector space and $\text{GL}(V)$ denotes the group of K -linear automorphisms of V . The *degree* (or *dimension*) of the representation is $\dim_K V$.

By choosing a basis of V we obtain an isomorphism $\text{GL}(V) \simeq \text{GL}_{\dim V}(K)$, recovering the matrix formulation. The two definitions are therefore equivalent.

Notation 1.8. Instead of $\rho(g)(v)$ we often write $\rho_g(v)$, or simply gv when ρ is clear from context.

1.3 Permutation Representations

Every group action on a set gives rise to a linear representation in a canonical way.

Definition 1.9 (Permutation representation). Let $G \curvearrowright X$ be a group action. Define the K -vector space

$$K[X] = \left\{ \sum_{x \in X}^{\text{finite}} \lambda_x e_x \mid \lambda_x \in K \right\}$$

with formal basis $\{e_x \mid x \in X\}$ indexed by X . The formula

$$g \cdot e_x := e_{g \cdot x}, \quad g \in G, x \in X,$$

extended linearly to all of $K[X]$, defines a representation

$$\rho: G \longrightarrow \text{GL}(K[X]).$$

This is called the *permutation representation* attached to $G \curvearrowright X$. Its degree is $\#X$.

Example 1.10. Let $G = S_3$ act on $X = \{1, 2, 3\}$ in the natural way. For $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$ in G , we have

$$\rho(\sigma) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \rho(\tau) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Indeed, σ sends $e_1 \mapsto e_2$, $e_2 \mapsto e_3$, $e_3 \mapsto e_1$, giving the first matrix (columns are images of basis vectors).

Definition 1.11 (Regular representation). In the special case where $X = G$ with G acting on itself by left multiplication $g \cdot x := gx$, the corresponding permutation representation $\rho: G \rightarrow \text{GL}(K[G])$ is called the *regular representation* of G over K .

The regular representation has degree $\#G$ and plays a central role in the theory: it contains every irreducible representation as a subrepresentation (a fact we will establish later).

1.4 Morphisms of Representations

Just as one studies groups via group homomorphisms, one studies representations via structure-preserving maps between them.

Equivalent representations

Let $\rho: G \rightarrow \text{GL}(V)$ be a representation of degree n over K . Picking a basis $\mathcal{B}_1$ of V yields a matrix representation $\rho_1: G \rightarrow \text{GL}_n(K)$; picking a different basis $\mathcal{B}_2$ yields $\rho_2: G \rightarrow \text{GL}_n(K)$. If $P \in \text{GL}_n(K)$ is the change-of-basis matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$, then

$$\rho_2(g) = P^{-1} \rho_1(g) P \quad \text{for all } g \in G.$$

Two representations related in this way are considered the same.

Example 1.12. Label the vertices of an equilateral triangle by 1, 2, 3. The symmetry group S_3 acts on the triangle, giving a representation $\rho: S_3 \rightarrow \text{GL}(\mathbb{R}^2)$: for $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$, $\rho(\sigma)$ is a rotation by $2\pi/3$ and $\rho(\tau)$ is a reflection. With respect to the standard basis of $\mathbb{R}^2$:

$$\rho_1: S_3 \rightarrow \text{GL}_2(\mathbb{R}), \quad \rho_1(\sigma) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \quad \rho_1(\tau) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

With a better-adapted basis one obtains the equivalent representation

$$\rho_2: S_3 \rightarrow \text{GL}_2(\mathbb{R}), \quad \rho_2(\sigma) = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \quad \rho_2(\tau) = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Definition 1.13 (Morphism of representations). Let G be a group, K a field, and $\rho_1: G \rightarrow \text{GL}(V_1)$, $\rho_2: G \rightarrow \text{GL}(V_2)$ two representations of G over K . A *morphism* from ρ_1 to ρ_2 is a K -linear map $T: V_1 \rightarrow V_2$ such that

$$T(\rho_1(g)(v)) = \rho_2(g)(T(v)) \quad \text{for all } g \in G, v \in V_1,$$

or equivalently, using the shorthand gv ,

$$T(gv) = gT(v).$$

In other words, the diagram

$$\begin{array}{ccc} V_1 & \xrightarrow{\rho_1(g)} & V_1 \\ T \downarrow & & \downarrow T \\ V_2 & \xrightarrow{\rho_2(g)} & V_2 \end{array}$$

commutes for every $g \in G$. An *isomorphism* of representations is a bijective morphism; isomorphic representations are also called *equivalent*. The set of all morphisms from ρ_1 to ρ_2 is a K -vector space denoted $\text{Hom}_G(V_1, V_2)$, or $\text{End}_G(V_1)$ when $V_1 = V_2$.

1.5 New Representations from Old

Definition 1.14 (Trivial representation). Let G be a group and K a field. Viewing $V = K$ as a one-dimensional vector space over itself, the *trivial representation* is the morphism

$$\mathbf{1}: G \longrightarrow \text{GL}(V), \quad g \mapsto \text{id}_V.$$

Every $g \in G$ acts as the identity.

Definition 1.15 (Direct sum). Let V and W be representations of G over K . Their *direct sum* $V \oplus W \simeq V \times W$ is again a representation of G , via

$$g(v, w) = (gv, gw) \quad (g \in G, v \in V, w \in W).$$

Definition 1.16 (Hom-representation and dual). Let V and W be representations of G over K . The space $\text{Hom}(V, W)$ of K -linear maps from V to W becomes a representation of G by

$$(gT)(v) = g(T(g^{-1}v)) \quad (g \in G, T \in \text{Hom}(V, W), v \in V).$$

The special case $V = W$ gives $\text{End}(V) = \text{Hom}(V, V)$. The special case $W = \mathbf{1}$ gives the *dual representation* $V^\vee = \text{Hom}(V, \mathbf{1})$, with G -action

$$(g\ell)(v) = \ell(g^{-1}v) \quad (g \in G, \ell \in V^\vee, v \in V).$$

Remark 1.17. A linear map $T \in \text{Hom}(V, W)$ is G -invariant (i.e. fixed by the G -action on $\text{Hom}(V, W)$) if and only if $T(gv) = gT(v)$ for all $g \in G$, i.e. if and only if T is a morphism of representations. Thus

$$\text{Hom}(V, W)^G = \text{Hom}_G(V, W).$$

Definition 1.18 (Tensor product of vector spaces). Let V and W be vector spaces over K . Their *tensor product* $V \otimes_K W$ is the K -vector space generated by symbols $v \otimes w$ ($v \in V, w \in W$) subject to the relations

$$\begin{aligned} (v + v') \otimes w &= v \otimes w + v' \otimes w, \\ v \otimes (w + w') &= v \otimes w + v \otimes w', \\ (kv) \otimes w &= v \otimes (kw) = k(v \otimes w), \end{aligned}$$

for all $v, v' \in V, w, w' \in W, k \in K$. Concretely, if $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W , then the nm symbols $v_i \otimes w_j$ form a basis of $V \otimes W$, so $\dim(V \otimes W) = \dim V \cdot \dim W$. A general element is a finite sum $\sum_k \tilde{v}_k \otimes \tilde{w}_k$ (not necessarily a pure tensor). Tensor product is associative and distributes over direct sums: $(U \otimes V) \otimes W \simeq U \otimes (V \otimes W)$ and $(U \oplus V) \otimes W \simeq (U \otimes W) \oplus (V \otimes W)$.

Definition 1.19 (Tensor product representation). Let V and W be representations of G over K . The vector space $V \otimes W$ becomes a representation of G via

$$g(v \otimes w) = (gv) \otimes (gw) \quad (g \in G, v \in V, w \in W),$$

extended linearly to all of $V \otimes W$. The associativity and distributivity of Definition 1.18 carry over to G -representations.

Remark 1.20 (Kronecker product). If g acts on V by the $n \times n$ matrix $A = (a_{ij})$ in a basis $v_1, \dots, v_n$, and on W by the $m \times m$ matrix $B = (b_{kl})$ in $w_1, \dots, w_m$, then g acts on $V \otimes W$ by the *Kronecker product*

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \cdots & a_{nn}B \end{pmatrix} \in \mathcal{M}_{nm}(K)$$

with respect to the basis $v_1 \otimes w_1, \dots, v_1 \otimes w_m, \dots, v_n \otimes w_1, \dots, v_n \otimes w_m$.

Proposition 1.21. For representations V and W of G there is a natural isomorphism of G -representations

$$V \otimes W \simeq \operatorname{Hom}(V^\vee, W).$$

Proof. The linear isomorphism $\phi: V \otimes W \xrightarrow{\sim} \operatorname{Hom}(V^\vee, W)$, $v \otimes w \mapsto (\ell \mapsto \ell(v)w)$, intertwines the G -actions: $\phi(g(v \otimes w)) = \phi(gv \otimes gw)$ is the map $\ell \mapsto \ell(gv)gw$, and $(g \cdot \phi(v \otimes w))(\ell) = g\phi(v \otimes w)(g^{-1}\ell) = g(\ell(gv)w) = \ell(gv)gw$. $\square$

1.6 Irreducibility and Indecomposability

Definition 1.22 (Subrepresentation). Let V be a representation of G . A *subrepresentation* of V is a subspace $W \subseteq V$ that is *stable* under G : $gw \in W$ for all $g \in G$ and $w \in W$.

Example 1.23. The subspace

$$V^G = \{v \in V \mid gv = v \text{ for all } g \in G\}$$

is a subrepresentation of V , called the space of G -invariants. It is isomorphic to a direct sum of copies of the trivial representation $\mathbf{1}$. Note also that $V^G = \operatorname{Hom}_G(\mathbf{1}, V)$.

Definition 1.24 (Irreducible representation). A representation V is *irreducible* (or *simple*) if its only subrepresentations are $\{0\}$ and V itself.

Example 1.25. Consider again $G = S_3 \curvearrowright X = \{1, 2, 3\}$ and the permutation representation $V = K[X]$. The vector $v = e_1 + e_2 + e_3$ is fixed by every $g \in G$, so $W = \operatorname{span}\{e_1 + e_2 + e_3\}$ is a nontrivial subrepresentation isomorphic to $\mathbf{1}$. Hence V is *not* irreducible.

Definition 1.26 (Indecomposable representation). A representation V is *indecomposable* if it is not isomorphic to a direct sum $V_1 \oplus V_2$ with $V_1, V_2 \neq \{0\}$.

Every irreducible representation is indecomposable, but the converse fails in general.

Example 1.27. Take $G = (K, +)$ with the representation $\rho: G \rightarrow \operatorname{GL}_2(K)$, $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ from Example 1.6. The subspace spanned by the first standard basis vector e_1 is stable, giving a subrepresentation isomorphic to $\mathbf{1}$, so ρ is *reducible*. However, ρ is *indecomposable*: $\operatorname{span}\{e_1\}$ has no G -stable complement in K^2 . Indeed, any complement would be one-dimensional, spanned by some $u = (a, b)^\top$ with $b \neq 0$. For each $x \in K$ stability requires $\rho(x)u = \mu u$ for some scalar $\mu \in K$; reading the second component gives $\mu = 1$, and the first component then yields $a + xb = a$, i.e. $xb = 0$ for all $x \in K$. Since $b \neq 0$ and K contains a non-zero element, this is impossible.

Effect on matrices

Let $\rho: G \rightarrow \operatorname{GL}_n(K)$ be a representation. In terms of matrices:

- ρ is *reducible* if and only if there exists $P \in \operatorname{GL}_n(K)$ such that all matrices $P^{-1}\rho(g)P$ are upper block-triangular of the form $\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$.

- ρ is *decomposable* if and only if there exists $P \in \mathrm{GL}_n(K)$ such that all matrices $P^{-1}\rho(g)P$ are block-diagonal of the form $\begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix}$.

Example 1.28. Return once more to the permutation representation $V = K[X]$ of $G = S_3 \curvearrowright X = \{1, 2, 3\}$. We have seen that $W = \mathrm{span}\{e_1 + e_2 + e_3\} \simeq \mathbf{1}$ is a subrepresentation. Consider its candidate complement

$$W' = \left\{ \sum_{x \in X} \lambda_x e_x \in V \mid \sum_{x \in X} \lambda_x = 0 \right\}.$$

One checks that W' is also a subrepresentation. If $3 \neq 0$ in K , then $V = W \oplus W'$ and V is decomposable. But if $K = \mathbb{Z}/3\mathbb{Z}$, then $e_1 + e_2 + e_3 \in W'$ (since $1 + 1 + 1 = 0$ in K), so $W \subset W'$ and the decomposition fails; in this case V is indecomposable.

Chapter 2

The Module Point of View

The language of modules over a ring provides a unified framework that subsumes both the theory of vector spaces and the theory of group representations. Recasting representations as modules over the group ring reveals structural properties — subrepresentations, morphisms, semi-simplicity — as instances of general module-theoretic phenomena, and leads directly to the fundamental theorem of this chapter: Maschke's theorem.

2.1 Modules over a Ring

Definition 2.1 (Module, morphism of modules). Let R be a ring. An R -module is a set M equipped with two operations

$$M \times M \rightarrow M, \quad (m, n) \mapsto m + n, \quad R \times M \rightarrow M, \quad (\lambda, m) \mapsto \lambda m,$$

such that $(M, +)$ is an abelian group, and for all $\lambda, \mu \in R$ and $m, n \in M$:

$$\lambda(\mu m) = (\lambda\mu)m, \quad 1m = m, \quad (\lambda + \mu)m = (\lambda m) + (\mu m), \quad \lambda(m + n) = (\lambda m) + (\lambda n).$$

A morphism $f: M \rightarrow N$ of R -modules is an R -linear map, meaning

$$f(m + m') = f(m) + f(m') \quad \text{and} \quad f(\lambda m) = \lambda f(m)$$

for all $m, m' \in M$ and $\lambda \in R$.

Example 2.2. • If R is a field, then an R -module is exactly an R -vector space, and a morphism of R -modules is exactly a linear map.

- For any ring R and $n \in \mathbb{N}$, the set $R^n = \{(x_1, \dots, x_n) \mid x_i \in R\}$ is an R -module under componentwise addition and scalar multiplication.
- A $\mathbb{Z}$ -module is exactly an abelian group: the scalar multiplication is given by $ng = g + \dots + g$ (n times) for $n \in \mathbb{Z}$, $g \in G$, and a morphism of $\mathbb{Z}$ -modules is exactly a group homomorphism.

Notation 2.3. Let R be a ring. Given two R -modules M and N , the set of morphisms from M to N is denoted $\text{Hom}_R(M, N)$; it is an abelian group under pointwise addition. In the case $M = N$ we write $\text{End}_R(M) := \text{Hom}_R(M, M)$, which is in fact a ring under composition. If R is a field and M is an R -vector space of dimension n , then $\text{End}_R(M) \simeq \mathcal{M}_n(R)$, the ring of $n \times n$ matrices over R .

Definition 2.4 (Submodule). Let M be an R -module. A *submodule* of M is a nonempty subset $N \subseteq M$ that is closed under addition and under multiplication by R .

Example 2.5. Viewing R as an R -module over itself, its submodules are precisely its *ideals*.

Proposition 2.6. Let $f: M \rightarrow N$ be a morphism of R -modules. Then $\text{Ker } f$ is a submodule of M , and $\text{Im } f$ is a submodule of N .

Proof. $\text{Ker } f$ is non-empty since $f(0) = 0$. For $m, m' \in \text{Ker } f$ and $\lambda \in R$, $f(m + m') = f(m) + f(m') = 0$ and $f(\lambda m) = \lambda f(m) = 0$, so $\text{Ker } f$ is closed under addition and scalar multiplication.

$\text{Im } f$ is non-empty since $0 = f(0) \in \text{Im } f$. For $n = f(m)$ and $n' = f(m')$ in $\text{Im } f$ and $\lambda \in R$: $n + n' = f(m + m') \in \text{Im } f$ and $\lambda n = f(\lambda m) \in \text{Im } f$. $\square$

Definition 2.7 (Direct sum). Let M be an R -module and $(M_i)_{i \in I}$ a collection of submodules. We say that $M = \bigoplus_{i \in I} M_i$ if every element $m \in M$ can be expressed as $m = \sum_{i \in I}^{\text{finite}} m_i$ with $m_i \in M_i$ in a *unique* way.

Example 2.8. The polynomial ring $R[x] = \bigoplus_{n \geq 0} Rx^n$ as an R -module.

Remark 2.9. For two submodules, $M = M_1 \oplus M_2$ if and only if every $m \in M$ is uniquely of the form $m = m_1 + m_2$ with $m_1 \in M_1$ and $m_2 \in M_2$, which is equivalent to $M_1 + M_2 = M$ and $M_1 \cap M_2 = \{0\}$. Indeed, if $m_1 + m_2 = m'_1 + m'_2$, then $m'_1 - m_1 = m_2 - m'_2 \in M_1 \cap M_2$.

2.2 Representations as Modules

The group ring

Definition 2.10 (Group ring). Let K be a field and G a group. Recall that

$$K[G] = \left\{ \sum_{g \in G}^{\text{finite}} \lambda_g e_g \mid \lambda_g \in K \right\} = \bigoplus_{g \in G} K e_g$$

is the K -vector space with basis $\{e_g\}_{g \in G}$. When $X = G$ is a group, $K[G]$ becomes a *ring* — the *group ring* of G over K — under the multiplication rule $e_g \cdot e_h := e_{gh}$, extended bilinearly:

$$\left(\sum_{g \in G} \lambda_g e_g \right) \left(\sum_{g \in G} \mu_g e_g \right) = \sum_{g \in G} \left(\sum_{\substack{g_1, g_2 \in G \\ g_1 g_2 = g}} \lambda_{g_1} \mu_{g_2} \right) e_g.$$

Remark 2.11. The ring $K[G]$ is commutative if and only if G is abelian. It contains a copy of K as the subring $\{\lambda e_{1_G} \mid \lambda \in K\}$.

$K[G]$ -modules and representations

The key observation is that $K[G]$ -modules are exactly representations of G over K .

More precisely, let M be a $K[G]$ -module. Then M is a K -vector space via $\lambda m := (\lambda e_{1_G})m$ for $\lambda \in K, m \in M$. The representation is then

$$\rho: G \rightarrow \mathrm{GL}(M), \quad g \mapsto (m \mapsto e_g \cdot m).$$

Conversely, given a K -vector space V and a representation $\rho: G \rightarrow \mathrm{GL}(V)$, we make V into a $K[G]$ -module by

$$\left(\sum_{g \in G} \lambda_g e_g \right) v := \sum_{g \in G} \lambda_g \rho(g)(v) \quad (\lambda_g \in K, v \in V).$$

These two constructions are inverse to each other. Under this correspondence:

- sub- $K[G]$ -modules = subrepresentations,
- $K[G]$ -module morphisms = morphisms of representations.

In particular, if $f: V \rightarrow W$ is a morphism of representations of G , then $\mathrm{Ker} f \subseteq V$ and $\mathrm{Im} f \subseteq W$ are subrepresentations. Moreover, irreducible representations correspond to *simple* $K[G]$ -modules, and indecomposable representations to *indecomposable* $K[G]$ -modules.

Definition 2.12 (Simple and indecomposable modules). Let R be a ring and M an R -module.

- M is *simple* if $M \neq \{0\}$ and its only submodules are $\{0\}$ and M .
- M is *indecomposable* if it cannot be expressed non-trivially as $M = \bigoplus_{i \in I} M_i$ with all $M_i \neq \{0\}$.

2.3 Semi-simplicity

Definition 2.13 (Semi-simple module). Let R be a ring. An R -module M is *semi-simple* if it can be decomposed as

$$M = \bigoplus_{i \in I} M_i$$

with each M_i a simple R -submodule. Since the M_i are simple, this is a “complete decomposition”. A representation is semi-simple if and only if the corresponding $K[G]$ -module is semi-simple, i.e., it is completely decomposable into irreducibles.

Definition 2.14 (Supplement). Let R be a ring, M an R -module, and $N \subseteq M$ a submodule. A *supplement* to N is a submodule $N' \subseteq M$ such that $M = N \oplus N'$.

Remark 2.15. In general, a supplement need not exist. For example, take $R = \mathbb{Z}$, $M = \mathbb{Z}$, $N = 2\mathbb{Z}$. If $\mathbb{Z} = 2\mathbb{Z} \oplus N'$, then for any $n' \in N'$ we have $2n' \in N' \cap 2\mathbb{Z} = \{0\}$, so $N' = \{0\}$, but then $2\mathbb{Z} \oplus \{0\} \neq \mathbb{Z}$.

Theorem 2.16. Let R be a ring and M an R -module. Assume there is no infinite strictly descending chain $M \supsetneq M_0 \supsetneq M_1 \supsetneq M_2 \supsetneq \cdots$ of submodules. Then

$$M \text{ is semi-simple} \iff \text{every submodule of } M \text{ has a supplement.}$$

Proof. ($\Leftarrow$): If M is simple, there is nothing to prove. Otherwise let $\{0\} \subsetneq N \subsetneq M$ be a proper nonzero submodule; by hypothesis it has a supplement N' , so $M = N \oplus N'$. We now iterate: apply the same argument to N and N' separately. This process terminates, since at each step the submodules strictly decrease, and by assumption there is no infinite descending chain.

($\Rightarrow$): Write $M = \bigoplus_{i \in I} M_i$ with each M_i simple. The index set I must be finite, else we would have an infinite strictly descending chain. Let $N \subseteq M$ be any submodule. Let $J \subseteq I$ be a maximal subset such that $M' := N \oplus \bigoplus_{i \in J} M_i$ is a direct sum. We claim $M' = M$, so that $\bigoplus_{i \in J} M_i$ is a supplement to N . Indeed, if $M_i \not\subseteq M'$ for some $i \in I$, then since M_i is simple we have $M' \cap M_i = \{0\}$, so we could add i to J and still have a direct sum, contradicting maximality. $\square$

2.4 Maschke's Theorem

We first recall a convenient characterisation of direct sum decompositions in terms of projections.

Theorem 2.17 (Projections). *Let K be a field and V a K -vector space. If $V = V_1 \oplus V_2$, the projection onto V_1 parallel to V_2 is the map*

$$\pi: V \rightarrow V, \quad v_1 + v_2 \mapsto v_1.$$

It is linear and satisfies $\text{Im } \pi = V_1$, $\text{Ker } \pi = V_2$, and $\pi^2 = \pi$. Conversely, if $\pi \in \text{End}_K(V)$ satisfies $\pi^2 = \pi$, then $V = \text{Im } \pi \oplus \text{Ker } \pi$, and π is the projection onto $\text{Im } \pi$ parallel to $\text{Ker } \pi$.

Proof. The forward direction is routine. For the converse, suppose $\pi^2 = \pi$. For any $v \in V$, write $v = \pi(v) + (v - \pi(v))$. Clearly $\pi(v) \in \text{Im } \pi$, and $\pi(v - \pi(v)) = \pi(v) - \pi^2(v) = \pi(v) - \pi(v) = 0$, so $v - \pi(v) \in \text{Ker } \pi$. Thus $V = \text{Im } \pi + \text{Ker } \pi$. If $v \in \text{Im } \pi \cap \text{Ker } \pi$, write $v = \pi(w)$; then $v = \pi(w) = \pi^2(w) = \pi(\pi(w)) = \pi(v) = 0$. So $\text{Im } \pi \cap \text{Ker } \pi = \{0\}$ and $V = \text{Im } \pi \oplus \text{Ker } \pi$. Finally, π acts as the identity on $\text{Im } \pi$ (just shown) and as zero on $\text{Ker } \pi$, so it is the projection onto $\text{Im } \pi$ parallel to $\text{Ker } \pi$. $\square$

Theorem 2.18 (Maschke). *Let K be a field and G a finite group of order $n = \#G$. If $n \neq 0$ in K (i.e. $\text{char } K \nmid n$), then every representation of G over K is semi-simple.*

Example 2.19. The permutation representation $V = K[\{1, 2, 3\}]$ induced by $S_3 \curvearrowright \{1, 2, 3\}$ is not semi-simple over $K = \mathbb{Z}/3\mathbb{Z}$, since $3 = 0$ in K . The subspace $W = \text{span}\{e_1 + e_2 + e_3\}$ is a subrepresentation isomorphic to $\mathbf{1}$. We claim W has no G -stable complement. Suppose otherwise; then there is a G -equivariant projection $\pi: V \rightarrow W$ with image W . By G -equivariance (since S_3 acts transitively on $\{e_1, e_2, e_3\}$), $\pi(e_1) = \pi(e_2) = \pi(e_3) = \alpha(e_1 + e_2 + e_3)$ for some $\alpha \in K$. For π to fix W we need $\pi(e_1 + e_2 + e_3) = e_1 + e_2 + e_3$, i.e. $3\alpha(e_1 + e_2 + e_3) = e_1 + e_2 + e_3$. But $3\alpha = 0$ in K for any α , contradicting $1 \neq 0$. This shows the hypothesis $n \neq 0$ in K is necessary.

Remark 2.20. The decomposition is in general not unique. For example, if V has degree $n \geq 2$ and G acts trivially, then $V \simeq \mathbf{1} \oplus \cdots \oplus \mathbf{1}$ (n copies), but the splitting into lines depends on the choice of basis.

Proof of Theorem 2.18. By Theorem 2.16, it suffices to show that every subrepresentation has a supplement. Since $\dim V < \infty$, there are no infinite descending chains of subrepresentations, so the hypothesis of Theorem 2.16 is satisfied.

Let $V_1 \subseteq V$ be a subrepresentation. As K -vector spaces (ignoring the G -action), we can write $V = V_1 \oplus V_2$ for some subspace V_2 ; this need not be G -stable. Let $\pi: V \rightarrow V$ be the projection onto V_1 parallel to V_2 (which exists by Theorem 2.17). Define the *Reynolds operator*

$$\Pi: V \rightarrow V, \quad v \mapsto \frac{1}{\#G} \sum_{g \in G} g \pi(g^{-1}v).$$

This is well-defined since $\#G \neq 0$ in K . We verify three properties.

$\text{Im } \Pi \subseteq V_1$ and $\Pi|_{V_1} = \text{id}_{V_1}$. Since V_1 is a subrepresentation, $g^{-1}v \in V_1$ whenever $v \in V_1$, so $\pi(g^{-1}v) = g^{-1}v$, hence $g\pi(g^{-1}v) = v$, giving $\Pi(v) = v$. For general v , each term $g\pi(g^{-1}v)$ lies in $gV_1 = V_1$, so $\Pi(v) \in V_1$. Thus $\text{Im } \Pi = V_1$ and $\Pi^2 = \Pi$.

Π is a morphism of representations. For any $h \in G$ and $v \in V$:

$$h \Pi(v) = \frac{1}{\#G} \sum_{g \in G} hg \pi(g^{-1}v) = \frac{1}{\#G} \sum_{g \in G} hg \pi((hg)^{-1}(hv)) = \frac{1}{\#G} \sum_{g \in G} g \pi(g^{-1}(hv)) = \Pi(hv),$$

where the last step uses that $g \mapsto hg$ is a bijection $G \rightarrow G$.

Conclusion. By Theorem 2.17, $\Pi^2 = \Pi$ implies $V = \text{Im } \Pi \oplus \text{Ker } \Pi = V_1 \oplus \text{Ker } \Pi$. Since Π is a morphism of representations, $\text{Ker } \Pi$ is a subrepresentation, and it is a supplement to V_1 . $\square$

Chapter 3

Character Theory

The character of a representation is a function $G \rightarrow \mathbb{C}$ that encodes an enormous amount of information about the representation in a highly compressed form. In this chapter we prove Schur's lemma, develop the inner-product structure on class functions, and show that irreducible characters form an orthonormal set that completely determines representations up to isomorphism.

Throughout this chapter and the rest of the notes, G denotes a finite group and $K = \mathbb{C}$. By Maschke's theorem, every finite-dimensional representation of G over $\mathbb{C}$ is semi-simple.

3.1 Schur's Lemma

Theorem 3.1 (Schur's lemma). *Let R be a ring, and let M_1, M_2 be simple R -modules. Then any module morphism $f: M_1 \rightarrow M_2$ is either 0 or an isomorphism.*

Proof. Since M_1 is simple, $\text{Ker } f$ is either $\{0\}$ or M_1 . If $\text{Ker } f = M_1$ then $f = 0$. Otherwise f is injective and $\text{Im } f \subseteq M_2$ is a nonzero submodule; since M_2 is simple, $\text{Im } f = M_2$, so f is an isomorphism. $\square$

Corollary 3.2. *If M is a simple R -module, then $\text{End}_R(M)$ is a division ring.*

Proof. Every nonzero endomorphism is an isomorphism by Theorem 3.1, hence invertible. $\square$

The decisive consequence for representations over $\mathbb{C}$ follows from the fact that $\mathbb{C}$ is algebraically closed.

Corollary 3.3 (Schur's lemma over $\mathbb{C}$). *Let G be a group, and let V, W be irreducible representations of G over $\mathbb{C}$. Then*

$$\text{Hom}_G(V, W) = \begin{cases} \{0\} & \text{if } V \not\cong W, \\ \{\lambda \text{ id}_V \mid \lambda \in \mathbb{C}\} & \text{if } V = W. \end{cases}$$

Proof. The case $V \not\cong W$ follows directly from Theorem 3.1. For $V = W$, let $f \in \text{End}_G(V)$. Since $\mathbb{C}$ is algebraically closed, f has at least one eigenvalue $\lambda \in \mathbb{C}$. Then $f - \lambda \text{ id} \in \text{End}_G(V)$ is not injective, so by Theorem 3.1 it must be 0, giving $f = \lambda \text{ id}$. $\square$

3.2 Dot Products

Definition 3.4 (Dot product over $\mathbb{R}$). Let V be an $\mathbb{R}$ -vector space. A *dot product* on V is a map $(v, w) \mapsto (v \mid w)$ from $V \times V$ to $\mathbb{R}$ that is bilinear, symmetric: $(v \mid w) = (w \mid v)$, and positive definite: $(v \mid v) \geq 0$ with equality only when $v = 0$.

Example 3.5. On $V = \mathbb{R}^n$, the usual dot product $(v \mid w) = \sum_{k=1}^n v_k w_k$. On $V = \mathbb{R}[x]$, one may take $(P \mid Q) = \int_0^1 P(x)Q(x) dx$.

Definition 3.6 (Dot product over $\mathbb{C}$). Let V be a $\mathbb{C}$ -vector space. A *dot product* (Hermitian inner product) on V is a map $(v, w) \mapsto (v \mid w)$ from $V \times V$ to $\mathbb{C}$ that is sesquilinear (linear in the first argument, conjugate-linear in the second: $(v \mid \lambda w) = \overline{\lambda}(v \mid w)$), conjugate-symmetric: $(v \mid w) = \overline{(w \mid v)}$, and positive definite: $(v \mid v) \in \mathbb{R}_{\geq 0}$ with equality only when $v = 0$.

Example 3.7. On $V = \mathbb{C}^n$, the standard product $(v \mid w) = \sum_{k=1}^n v_k \overline{w_k}$. On $\mathbb{C}[x]$, one may take $(P \mid Q) = \int_0^1 P(x) \overline{Q(x)} dx$.

Definition 3.8 (Orthogonal, orthonormal). A family $(v_j)_{j \in J}$ in V is *orthogonal* if $(v_j \mid v_k) = 0$ for all $j \neq k$, and *orthonormal* if additionally $(v_j \mid v_j) = 1$ for all j .

Proposition 3.9. If $(e_j)_{j \in J}$ is an orthonormal basis of V , then the coordinates of any $v \in V$ are recovered as $(v \mid e_j)$.

Proof. Write $v = \sum_{j \in J} \lambda_j e_j$. Then $(v \mid e_k) = \sum_{j \in J} \lambda_j (e_j \mid e_k) = \lambda_k$. $\square$

3.3 Conjugacy and Class Functions

Definition 3.10 (Conjugacy). Two elements $g, g' \in G$ are *conjugate* if $g' = hgh^{-1}$ for some $h \in G$. The *conjugacy class* of g is the set of all its conjugates.

Definition 3.11 (Class function). A *class function* is a function $\psi: G \rightarrow \mathbb{C}$ satisfying $\psi(hgh^{-1}) = \psi(g)$ for all $g, h \in G$, i.e., a function constant on conjugacy classes.

Class functions form a $\mathbb{C}$ -vector space of dimension equal to the number of conjugacy classes. We equip it with the inner product

$$(\phi \mid \psi) = \frac{1}{\#G} \sum_{g \in G} \phi(g) \overline{\psi(g)}.$$

3.4 Traces

Definition 3.12 (Trace). The *trace* of a square matrix A is the sum of its diagonal entries: $\text{tr } A = \sum_i A_{ii}$.

Example 3.13. $\operatorname{tr} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1 + 4 = 5.$

Proposition 3.14. *The trace is linear: $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$ and $\operatorname{tr}(\lambda A) = \lambda \operatorname{tr} A$. Furthermore $\operatorname{tr}(AB) = \operatorname{tr}(BA)$, whence $\operatorname{tr}(BAB^{-1}) = \operatorname{tr} A$.*

Corollary 3.15. *The trace of a linear map $T: V \rightarrow V$ is well-defined independently of the choice of basis: for any basis $(e_1, \dots, e_d)$ of V ,*

$$\operatorname{tr} T = \sum_i (\text{coefficient of } e_i \text{ in } T(e_i)).$$

3.5 The Character of a Representation

Definition 3.16 (Character). The *character* of a representation $\rho: G \rightarrow \operatorname{GL}(V)$ is the function

$$\chi_\rho: G \longrightarrow \mathbb{C}, \quad g \longmapsto \operatorname{tr} \rho(g).$$

Proposition 3.17. *The character of a representation is a class function.*

Proof. $\operatorname{tr} \rho(hgh^{-1}) = \operatorname{tr}(\rho(h)\rho(g)\rho(h)^{-1}) = \operatorname{tr} \rho(g).$ □

Proposition 3.18. *Equivalent representations have the same character.*

Proof. $\operatorname{tr}(P^{-1}\rho(g)P) = \operatorname{tr} \rho(g).$ □

Proposition 3.19. *The character of the trivial representation $\mathbf{1}$ is the constant function $\mathbf{1}: G \rightarrow \mathbb{C}$.*

Proof. By definition, $\mathbf{1}: G \rightarrow \operatorname{GL}(\mathbb{C})$ sends every g to $\operatorname{id}_{\mathbb{C}}$, so $\chi_{\mathbf{1}}(g) = \operatorname{tr}(\operatorname{id}_{\mathbb{C}}) = 1$ for all g . □

Proposition 3.20. *If χ is the character of a representation of degree n , then $\chi(1_G) = n$. We write $\deg \chi := \chi(1_G)$.*

Proof. Since ρ is a group morphism, $\rho(1_G) = \operatorname{id}_V$. Hence $\chi(1_G) = \operatorname{tr}(\operatorname{id}_V) = \dim V = n$. □

Proposition 3.21. *If V_1, V_2 are representations of G with characters χ_1, χ_2 , then the character of $V_1 \oplus V_2$ is $\chi_1 + \chi_2$.*

Proof. The action of g on $V_1 \oplus V_2$ is given by the block matrix $\begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}$, whose trace is $\operatorname{tr} \rho_1(g) + \operatorname{tr} \rho_2(g)$. □

3.6 Characters on Inverses and Kernels

Since G is finite, every element has finite order, which forces the eigenvalues of $\rho(g)$ to be roots of unity.

Lemma 3.22. *Let ρ be a representation of a finite group G of order $n = \#G$. For all $g \in G$, the matrix $\rho(g)$ is diagonalisable over $\mathbb{C}$, and its eigenvalues are of the form $e^{2k\pi i/n}$, $k \in \mathbb{Z}$.*

Proof. By Lagrange's theorem $g^n = 1_G$, so $\rho(g)^n = \text{id}$. The polynomial $P(x) = x^n - 1 = \prod_{k=0}^{n-1} (x - e^{2k\pi i/n})$ satisfies $P(\rho(g)) = 0$, has all roots in $\mathbb{C}$, and has no repeated roots. Hence $\rho(g)$ is diagonalisable with eigenvalues among the n -th roots of unity. $\square$

Corollary 3.23. *Let χ be the character of ρ . For all $g \in G$, $g \in \text{Ker } \rho \iff \chi(g) = \deg \rho$.*

Proof. Choose a basis diagonalising $\rho(g)$, with diagonal entries λ_j satisfying $|\lambda_j| = 1$. Then $\chi(g) = \sum_j \lambda_j$. This sum equals $n = \deg \rho$ if and only if all $\lambda_j = 1$, i.e., $\rho(g) = \text{id}$. $\square$

Corollary 3.24. *For all $g \in G$, $\chi(g^{-1}) = \overline{\chi(g)}$.*

Proof. With $\rho(g) = \text{diag}(\lambda_1, \dots, \lambda_n)$ as above, we have $\rho(g^{-1}) = \rho(g)^{-1} = \text{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1}) = \text{diag}(\overline{\lambda_1}, \dots, \overline{\lambda_n})$, so $\chi(g^{-1}) = \sum_j \overline{\lambda_j} = \overline{\chi(g)}$. $\square$

3.7 Invariants and the Character of Hom

Recall from Definition 2.7 that the space of G -invariants of V is $V^G = \{v \in V \mid gv = v \text{ for all } g \in G\} = \bigcap_{g \in G} \text{Ker}(\rho(g) - \text{id})$. Recall also that $\text{Hom}(V, W)^G = \text{Hom}_G(V, W)$.

Lemma 3.25. *Let $\rho: G \rightarrow \text{GL}(V)$ be a representation. The operator $\pi = \frac{1}{\#G} \sum_{g \in G} \rho(g)$ is a projection onto V^G .*

Proof. For any $v \in V$ and $h \in G$:

$$\rho(h)(\pi(v)) = \frac{1}{\#G} \sum_{g \in G} \rho(hg)(v) = \frac{1}{\#G} \sum_{g \in G} \rho(g)(v) = \pi(v),$$

so $\text{Im } \pi \subseteq V^G$. If $v \in V^G$ then $\rho(g)(v) = v$ for all g , giving $\pi(v) = v$. Thus π is the identity on V^G and maps all of V into V^G , so $\pi^2 = \pi$. $\square$

Corollary 3.26. $\dim V^G = \frac{1}{\#G} \sum_{g \in G} \chi(g)$.

Proof. In a suitable basis, π has matrix $\begin{pmatrix} I_n & 0 \\ 0 & 0 \end{pmatrix}$ with $n = \dim V^G$. Hence $\dim V^G = \text{tr } \pi = \frac{1}{\#G} \sum_g \text{tr } \rho(g)$. $\square$

Lemma 3.27. *The character of $\text{Hom}(V_1, V_2)$ is $\overline{\chi_1} \chi_2$, where χ_i is the character of V_i .*

Proof. Fix bases $(e_1, \dots, e_m)$ of V_1 and $(f_1, \dots, f_n)$ of V_2 . For each pair (i, j) with $1 \leq i \leq n$, $1 \leq j \leq m$, let $T_{ij} \in \text{Hom}(V_1, V_2)$ be the map $e_j \mapsto f_i$ sending the remaining basis vectors of V_1 to 0. These T_{ij} form a basis of $\text{Hom}(V_1, V_2)$. For $g \in G$ we have $gT_{ij} = \rho_2(g) T_{ij} \rho_1(g)^{-1}$. Writing $[\rho_1(g)^{-1}]_{kl}$ and $[\rho_2(g)]_{kl}$ for the matrix entries in the chosen bases,

$$(gT_{ij})(e_l) = \rho_2(g) T_{ij} \left(\sum_k [\rho_1(g)^{-1}]_{kl} e_k \right) = [\rho_1(g)^{-1}]_{jl} \rho_2(g) f_i = \sum_k [\rho_2(g)]_{ki} [\rho_1(g)^{-1}]_{jl} f_k,$$

so $gT_{ij} = \sum_{k,l} [\rho_2(g)]_{ki} [\rho_1(g)^{-1}]_{jl} T_{kl}$. The coefficient of T_{ij} in gT_{ij} is therefore $[\rho_2(g)]_{ii} [\rho_1(g)^{-1}]_{jj}$, and the trace is

$$\text{tr}(g \text{ on } \text{Hom}(V_1, V_2)) = \sum_{i,j} [\rho_2(g)]_{ii} [\rho_1(g)^{-1}]_{jj} = \chi_2(g) \chi_1(g^{-1}) = \chi_2(g) \overline{\chi_1(g)},$$

where the last step uses Corollary 3.24. □

Corollary 3.28. *The character of the dual $V^\vee = \text{Hom}(V, \mathbf{1})$ is $\overline{\chi}$.*

Proof. Apply Lemma 3.27 with $V_1 = V$ and $V_2 = \mathbf{1}$: the character of $V^\vee$ is $\overline{\chi} \cdot 1 = \overline{\chi}$. □

Corollary 3.29. *Let V_1, V_2 be representations of G with characters χ_1, χ_2 . Then*

$$(\chi_1 \mid \chi_2) = \dim \text{Hom}_G(V_1, V_2) \in \mathbb{Z}_{\geq 0}.$$

Proof. The character of $\text{Hom}(V_1, V_2)$ is $\overline{\chi_1} \chi_2$, so by Corollary 3.26,

$$\dim \text{Hom}_G(V_1, V_2) = \dim \text{Hom}(V_1, V_2)^G = \frac{1}{\#G} \sum_{g \in G} \overline{\chi_1(g)} \chi_2(g) = (\chi_1 \mid \chi_2). \quad \square$$

3.8 Tensor Products and Symmetric Powers

Proposition 3.30 (Character of a tensor product). *Let V and W be representations of G with characters χ_V and χ_W . Then*

$$\chi_{V \otimes W} = \chi_V \cdot \chi_W \quad (\text{pointwise product}).$$

Proof. By Proposition 1.21, $V \otimes W \simeq \text{Hom}(V^\vee, W)$ as G -representations. Applying Lemma 3.27 and the fact that $\chi_{V^\vee} = \overline{\chi_V}$:

$$\chi_{V \otimes W} = \chi_{\text{Hom}(V^\vee, W)} = \overline{\chi_{V^\vee}} \chi_W = \overline{\overline{\chi_V}} \chi_W = \chi_V \chi_W. \quad \square$$

Remark 3.31. Alternatively: if g acts on V by A and on W by B , the Kronecker product formula (Remark 1.20) gives $\chi_{V \otimes W}(g) = \text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B) = \chi_V(g) \chi_W(g)$, since the diagonal entries of $A \otimes B$ are $a_{ii} b_{kk}$.

Definition 3.32 (Symmetric and alternating squares). Let V be a G -representation over $\mathbb{C}$. The transposition $\tau: V \otimes V \rightarrow V \otimes V$, $v \otimes w \mapsto w \otimes v$, satisfies $\tau^2 = \text{id}$ and commutes with the G -action, so its ± 1 eigenspaces are G -stable. They are called the *symmetric square* and the *alternating*

square (or exterior square) of V :

$$S^2V = \text{span}\{v \otimes w + w \otimes v : v, w \in V\}, \quad \Lambda^2V = \text{span}\{v \otimes w - w \otimes v : v, w \in V\}.$$

If $v_1, \dots, v_n$ is a basis of V then $\{v_i \otimes v_j + v_j \otimes v_i : i \leq j\}$ is a basis of S^2V and $\{v_i \otimes v_j - v_j \otimes v_i : i < j\}$ is a basis of Λ^2V , giving $\dim S^2V = \frac{n(n+1)}{2}$ and $\dim \Lambda^2V = \frac{n(n-1)}{2}$. Since the two eigenspaces span $V \otimes V$, we have $V \otimes V = S^2V \oplus \Lambda^2V$.

Proposition 3.33 (Characters of the symmetric and alternating squares). *Let V be a representation of G with character χ . For all $g \in G$:*

$$\chi_{S^2V}(g) = \frac{1}{2}(\chi(g)^2 + \chi(g^2)), \quad \chi_{\Lambda^2V}(g) = \frac{1}{2}(\chi(g)^2 - \chi(g^2)).$$

Proof. Since g has finite order, $\rho(g)^{|g|} = I$, so $\rho(g)$ is diagonalizable over $\mathbb{C}$ with eigenvalues $\alpha_1, \dots, \alpha_n$ (roots of unity). The element g acts on $v_i \otimes v_j \pm v_j \otimes v_i$ with eigenvalue $\alpha_i \alpha_j$. Using the standard identity $\sum_{i \leq j} \alpha_i \alpha_j = \frac{1}{2}[(\sum_i \alpha_i)^2 + \sum_i \alpha_i^2]$ and $\sum_{i < j} \alpha_i \alpha_j = \frac{1}{2}[(\sum_i \alpha_i)^2 - \sum_i \alpha_i^2]$, together with $\sum_i \alpha_i = \chi(g)$ and $\sum_i \alpha_i^2 = \chi(g^2)$ (eigenvalues of $\rho(g^2)$ are α_i^2), the formulas follow. $\square$

Remark 3.34. As a consistency check, $\chi_{S^2V}(g) + \chi_{\Lambda^2V}(g) = \chi(g)^2 = \chi_V(g)^2 = \chi_{V \otimes V}(g)$, confirming the decomposition $V \otimes V = S^2V \oplus \Lambda^2V$. The formulas are practical: given the character table of G , one can decompose S^2V and Λ^2V into irreducibles using the first orthogonality relations, without constructing the representation explicitly.

3.9 First Orthogonality and the Main Theorem

Theorem 3.35 (First orthogonality of characters). *The set $\text{Irr}(G)$ is orthonormal with respect to the inner product on class functions.*

Proof. Let $\chi_1, \chi_2 \in \text{Irr}(G)$ with corresponding irreducible representations V_1, V_2 . By Corollary 3.29 and Corollary 3.3,

$$(\chi_1 | \chi_2) = \dim \text{Hom}_G(V_1, V_2) = \begin{cases} 0 & \text{if } V_1 \not\simeq V_2, \\ 1 & \text{if } V_1 \simeq V_2. \end{cases} \quad \square$$

Theorem 3.36 (Characters know everything). *Let V be a representation of G of character χ , with irreducible decomposition $V \simeq W_1^{\oplus n_1} \oplus \dots \oplus W_r^{\oplus n_r}$ (pairwise non-isomorphic W_j with characters χ_j). Then $\chi = \sum_j n_j \chi_j$ and*

$$n_j = (\chi | \chi_j) \quad \text{and} \quad (\chi | \chi) = \sum_{j=1}^r n_j^2.$$

Proof. Additivity of characters on direct sums gives $\chi = \sum_j n_j \chi_j$. Taking the inner product with χ_k and using orthonormality yields $(\chi | \chi_k) = n_k$. Then $(\chi | \chi) = \sum_{j,k} n_j n_k (\chi_j | \chi_k) = \sum_j n_j^2$. $\square$

Corollary 3.37. *V is irreducible if and only if $(\chi | \chi) = 1$.*

Proof. By Theorem 3.36, $(\chi | \chi) = \sum_j n_j^2$ with $n_j \in \mathbb{Z}_{\geq 0}$. This equals 1 iff exactly one $n_j = 1$ and the rest vanish, i.e. $V \simeq W_j$ for a unique irreducible W_j . $\square$

Corollary 3.38. *The multiplicities n_k are uniquely determined by V .*

Proof. By Theorem 3.36, $n_k = (\chi | \chi_k)$, which depends only on the character χ of V (and not on a chosen decomposition). $\square$

Corollary 3.39. *Two representations of G are equivalent if and only if they have the same character.*

Proof. Equivalent representations have the same character (already noted). Conversely, if $\chi_V = \chi_{V'}$, then $n_k = (\chi_V | \chi_k) = (\chi_{V'} | \chi_k) = n'_k$ for each k , so V and V' have the same decomposition into irreducibles and are therefore isomorphic. $\square$

Corollary 3.40. $\#\text{Irr}(G) \leq \#\{\text{conjugacy classes of } G\}$.

Proof. The elements of $\text{Irr}(G)$ are orthonormal in the space of class functions, which has dimension equal to the number of conjugacy classes. An orthonormal set cannot exceed the dimension of its ambient space. (We will prove equality in Theorem 4.21.) $\square$

Example 3.41 (S_3). Take $G = S_3$ with $\#G = 6$ and three conjugacy classes: $\{1_G\}$, $\{(12), (13), (23)\}$, $\{(123), (132)\}$. We already know three irreducible representations: the trivial $\mathbf{1}$, the sign ε , and the standard $\trianglelefteq$ of degree 2. The permutation representation Perm (induced by $S_3 \curvearrowright \{1, 2, 3\}$) has character $\chi_{\text{Perm}}(g) = \#\text{Fix } g$:

	$\{1_G\}$	$\{(12), (13), (23)\}$	$\{(123), (132)\}$
$\mathbf{1}$	1	1	1
ε	1	-1	1
$\trianglelefteq$	2	0	-1
Perm	3	1	0

We compute $(\mathbf{1} | \mathbf{1}) = (\varepsilon | \varepsilon) = (\trianglelefteq | \trianglelefteq) = 1$, confirming irreducibility. Since $(\text{Perm} | \text{Perm}) = \frac{1}{6}(9 + 3 + 0) = 2$, Perm is not irreducible. Computing $(\text{Perm} | \mathbf{1}) = 1$, $(\text{Perm} | \varepsilon) = 0$, $(\text{Perm} | \trianglelefteq) = 1$ gives $\text{Perm} \simeq \mathbf{1} \oplus \trianglelefteq$.

Chapter 4

Character Tables

In this chapter we determine the precise number of irreducible representations of a finite group (it equals the number of conjugacy classes), prove the Wedderburn structure theorem for $\mathbb{C}[G]$, and use these tools to compute character tables explicitly.

4.1 Decomposition of the Regular Representation

Proposition 4.1. *Let G act on a finite set X . The character of the permutation representation $\mathbb{C}[X]$ is $\chi(g) = \# \text{Fix } g$, where $\text{Fix } g = \{x \in X \mid g \cdot x = x\}$.*

Proof. In the basis $\{e_x\}_{x \in X}$, the action sends $e_x \mapsto e_{g \cdot x}$, so the diagonal entry of g at position x is 1 if $g \cdot x = x$ and 0 otherwise. $\square$

Applying this to the regular representation $G \curvearrowright G$ by left multiplication:

$$\chi_{\text{reg}}(g) = \# \text{Fix } g = \#\{h \in G \mid gh = h\} = \begin{cases} \#G & \text{if } g = 1_G, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 4.2. The regular representation is *faithful*: if $g \cdot x = x$ for all $x \in G$, then $g \cdot 1_G = 1_G$ forces $g = 1_G$. Hence every finite group admits a faithful representation. In particular, a non-abelian group must have at least one irreducible representation of degree ≥ 2 .

Theorem 4.3 (Decomposition of the regular representation).

$$\mathbb{C}[G] \simeq \bigoplus_{\rho \in \text{Irr}(G)} \rho^{\oplus \deg \rho}.$$

Proof. For any $\rho \in \text{Irr}(G)$ with character χ_ρ , its multiplicity in $\mathbb{C}[G]$ is

$$(\chi_{\text{reg}} \mid \chi_\rho) = \frac{1}{\#G} \sum_{g \in G} \chi_{\text{reg}}(g) \overline{\chi_\rho(g)} = \frac{1}{\#G} \cdot \#G \cdot \chi_\rho(1_G) = \deg \rho. \quad \square$$

Corollary 4.4. $\#G = \sum_{\rho \in \text{Irr}(G)} (\deg \rho)^2$.

Proof. Compare dimensions on both sides of Theorem 4.3: $\#G = \sum_{\rho} (\deg \rho) \cdot \deg \rho$. $\square$

Example 4.5. For $G = S_3$ with $\text{Irr}(G) = \{1, \varepsilon, \trianglelefteq\}$: $\mathbb{C}[S_3] \simeq 1 \oplus \varepsilon \oplus \trianglelefteq^{\oplus 2}$, and $6 = 1^2 + 1^2 + 2^2$.
✓

Corollary 4.6. For all $g \in G$,

$$\sum_{\chi \in \text{Irr}(G)} (\deg \chi) \chi(g) = \begin{cases} \#G & \text{if } g = 1_G, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. This is the statement that the character of $\bigoplus_{\rho} \rho^{\oplus \deg \rho}$ equals χ_{reg} . $\square$

4.2 Wedderburn on $\mathbb{C}[G]$

Definition 4.7 (K -algebra). Let K be a field. A K -algebra is a K -vector space A which is also a ring satisfying $\lambda(ab) = (\lambda a)b = a(\lambda b)$ for all $\lambda \in K$ and $a, b \in A$.

Example 4.8. The polynomial ring $K[x]$, the matrix ring $\mathcal{M}_n(K)$, and the group ring $\mathbb{C}[G]$ are all K -algebras.

Definition 4.9 (Opposite ring). The *opposite ring* R^{opp} of a ring R has the same underlying set and addition, with multiplication reversed: $x \times_{R^{\text{opp}}} y = y \times_R x$.

Example 4.10. If R is commutative, $R^{\text{opp}} = R$. Transposition gives $\mathcal{M}_n(K)^{\text{opp}} \simeq \mathcal{M}_n(K)$.

Lemma 4.11. Let $M = R$ viewed as an R -module. Then $\text{End}_R(M) \simeq R^{\text{opp}}$.

Proof. The map $r \mapsto (\mu_r: m \mapsto mr)$ from R^{opp} to $\text{End}_R(M)$ is injective (since $\mu_r(1) = r$), surjective (any $f \in \text{End}_R(M)$ satisfies $f = \mu_{f(1)}$), and a ring morphism ($\mu_r \circ \mu_s = \mu_{sr}$). $\square$

Lemma 4.12 (Endomorphisms of direct sums). If $M = M_1 \oplus M_2$ is an R -module, then

$$\text{End}_R(M) \simeq \begin{pmatrix} \text{End}_R(M_1) & \text{Hom}_R(M_2, M_1) \\ \text{Hom}_R(M_1, M_2) & \text{End}_R(M_2) \end{pmatrix}.$$

If $M_1 \not\simeq M_2$ are both simple, this reduces to $\text{End}_R(M) \simeq \text{End}_R(M_1) \times \text{End}_R(M_2)$.

Proof. Let $\iota_i: M_i \hookrightarrow M$ be the canonical inclusions and $\pi_i: M \twoheadrightarrow M_i$ the projections, satisfying $\pi_i \iota_j = \delta_{ij} \text{id}_{M_i}$ and $\iota_1 \pi_1 + \iota_2 \pi_2 = \text{id}_M$. For $f \in \text{End}_R(M)$, define $f_{ij} = \pi_i f \iota_j \in \text{Hom}_R(M_j, M_i)$. The map $f \mapsto (f_{ij})$ is a K -linear bijection from $\text{End}_R(M)$ to the matrix space, with inverse $(f_{ij}) \mapsto \sum_{i,j} \iota_i f_{ij} \pi_j$. It is a ring morphism: if $f, g \in \text{End}_R(M)$, then

$$(fg)_{ij} = \pi_i f g \iota_j = \pi_i f (\iota_1 \pi_1 + \iota_2 \pi_2) g \iota_j = \sum_k f_{ik} g_{kj},$$

which is matrix multiplication. For the second statement: if $M_1 \not\simeq M_2$ are simple, then Schur's lemma (Theorem 3.1) gives $\text{Hom}_R(M_1, M_2) = \text{Hom}_R(M_2, M_1) = 0$, so the off-diagonal entries vanish and the matrix ring becomes the product $\text{End}_R(M_1) \times \text{End}_R(M_2)$. $\square$

Theorem 4.13 (Wedderburn–Artin). *Let K be a field, and let A be a K -algebra that is semi-simple as an A -module. Then*

$$A \simeq \prod_i \mathcal{M}_{n_i}(D_i),$$

where the D_i are division rings containing K .

Remark 4.14. We state Wedderburn–Artin as background context only: its proof in full generality requires the Jacobson density theorem and other tools from noncommutative algebra. For the case $A = \mathbb{C}[G]$ we will give a direct proof below, using only Maschke, Schur, and the decomposition of the regular representation.

Theorem 4.15 (Wedderburn decomposition of $\mathbb{C}[G]$).

$$\mathbb{C}[G] \simeq \prod_{\rho \in \text{Irr}(G)} \mathcal{M}_{\deg \rho}(\mathbb{C}).$$

Proof. By Lemma 4.11, $\mathbb{C}[G]^{\text{opp}} \simeq \text{End}_{\mathbb{C}[G]}(\mathbb{C}[G]) = \text{End}_G(\mathbb{C}[G])$. By Theorem 4.3, $\mathbb{C}[G] \simeq \bigoplus_{\rho \in \text{Irr}(G)} \rho^{\oplus \deg \rho}$ as G -modules. Iterated application of Lemma 4.12, together with the vanishing $\text{Hom}_G(\rho, \rho') = 0$ for $\rho \neq \rho'$ (Schur), gives

$$\text{End}_G(\mathbb{C}[G]) \simeq \prod_{\rho \in \text{Irr}(G)} \text{End}_G(\rho^{\oplus \deg \rho}) \simeq \prod_{\rho \in \text{Irr}(G)} \mathcal{M}_{\deg \rho}(\text{End}_G(\rho)).$$

By Corollary 3.3, $\text{End}_G(\rho) \simeq \mathbb{C}$ for every irreducible ρ over $\mathbb{C}$, so

$$\mathbb{C}[G]^{\text{opp}} \simeq \prod_{\rho \in \text{Irr}(G)} \mathcal{M}_{\deg \rho}(\mathbb{C}).$$

Taking opposites on both sides and using $\mathcal{M}_n(\mathbb{C})^{\text{opp}} \simeq \mathcal{M}_n(\mathbb{C})$ (via transposition) yields the claim. $\square$

Fourier inversion formula

The Wedderburn isomorphism is given concretely by $x \mapsto (\rho(x))_{\rho \in \text{Irr}(G)}$. Its inverse sends the element $(0, \dots, 0, \text{id}, 0, \dots, 0) \in \prod_{\rho} \mathcal{M}_{\deg \rho}(\mathbb{C})$ (identity in the ρ -slot, zero elsewhere) to the idempotent

$$\pi_{\rho} = \frac{\deg \rho}{\#G} \sum_{g \in G} \overline{\chi_{\rho}(g)} e_g \in \mathbb{C}[G].$$

This *Fourier inversion formula* provides the projections onto isotypic components, as we discuss in Section 4.4.

4.3 The Number of Irreducible Representations

Definition 4.16 (Centre). The *centre* of a ring R is $Z(R) = \{z \in R \mid zr = rz \text{ for all } r \in R\}$.

Theorem 4.17. For every field K and $n \in \mathbb{N}$, $Z(\mathcal{M}_n(K)) = \{\lambda I_n \mid \lambda \in K\} \simeq K$.

Proof. Clearly every scalar matrix λI_n commutes with all of $\mathcal{M}_n(K)$. Conversely, let $A = (a_{ij}) \in Z(\mathcal{M}_n(K))$ and let E_{kl} denote the matrix with a 1 in position (k, l) and zeros elsewhere. Then

$$[AE_{kl}]_{ij} = a_{ik} \delta_{jl}, \quad [E_{kl}A]_{ij} = \delta_{ik} a_{lj},$$

so $AE_{kl} = E_{kl}A$ for all k, l gives $a_{ik} \delta_{jl} = \delta_{ik} a_{lj}$ for all i, j, k, l . Setting $i \neq k$ and $j = l$ yields $a_{ik} = 0$: all off-diagonal entries of A vanish. Setting $i = k$ and $j = l$ gives $a_{kk} = a_{ll}$: all diagonal entries are equal. Hence $A = \lambda I_n$ for some $\lambda \in K$. The map $\lambda \mapsto \lambda I_n$ is a ring isomorphism $K \rightarrow Z(\mathcal{M}_n(K))$. $\square$

Proposition 4.18. *An element $x = \sum_{g \in G} \lambda_g e_g \in \mathbb{C}[G]$ lies in $Z(\mathbb{C}[G])$ if and only if the function $g \mapsto \lambda_g$ is a class function.*

Proof. $x \in Z(\mathbb{C}[G])$ iff $xe_h = e_h x$ for all $h \in G$, which amounts to $\lambda_g = \lambda_{hgh^{-1}}$ for all $g, h \in G$. $\square$

Corollary 4.19. *The class sums $e_C = \sum_{g \in C} e_g$, as C ranges over conjugacy classes, form a $\mathbb{C}$ -basis of $Z(\mathbb{C}[G])$. In particular, $\dim_{\mathbb{C}} Z(\mathbb{C}[G]) = \#\{\text{conjugacy classes in } G\}$.*

Proof. By Proposition 4.18, $x = \sum_g \lambda_g e_g \in Z(\mathbb{C}[G])$ iff the function $g \mapsto \lambda_g$ is constant on conjugacy classes. Thus every $x \in Z(\mathbb{C}[G])$ may be written uniquely as $x = \sum_C c_C e_C$ with $c_C \in \mathbb{C}$, so the class sums $\{e_C\}$ span $Z(\mathbb{C}[G])$. They are linearly independent because the $\{e_g\}_{g \in G}$ are a basis of $\mathbb{C}[G]$ and distinct conjugacy classes are disjoint. $\square$

Remark 4.20. If M is any R -module, then every $z \in Z(R)$ acts on M as an R -module endomorphism via $m \mapsto zm$. In particular, if $f: G \rightarrow \mathbb{C}$ is a class function, then $T_f = \sum_{g \in G} f(g)\rho(g)$ lies in $\text{End}_G(V)$ for every representation V .

Theorem 4.21 (Number of irreducible representations). *Let G be a finite group. Then*

$$\#\text{Irr}(G) = \#\{\text{conjugacy classes in } G\}.$$

Proof. Using Theorem 4.15 and Theorem 4.17:

$$\#\{\text{conj. classes}\} = \dim Z(\mathbb{C}[G]) = \dim Z\left(\prod_{\rho \in \text{Irr } G} \mathcal{M}_{\deg \rho}(\mathbb{C})\right) = \sum_{\rho \in \text{Irr } G} \dim Z(\mathcal{M}_{\deg \rho}(\mathbb{C})) = \#\text{Irr}(G).$$

$\square$

Corollary 4.22. *The irreducible characters form an orthonormal basis of the space of class functions.*

Proof. They are orthonormal by Theorem 3.35. The space of class functions has dimension equal to the number of conjugacy classes, which by Theorem 4.21 equals $\#\text{Irr}(G)$. An orthonormal set of the right cardinality is a basis. $\square$

Corollary 4.23. *G is abelian if and only if all irreducible representations of G have degree 1.*

Proof. G abelian $\iff$ every conjugacy class is a singleton $\iff \#\text{Irr}(G) = \#G$. Combined with $\sum_{\rho} (\deg \rho)^2 = \#G$ (Corollary 4.4), this gives $\sum_{\rho} (\deg \rho)^2 = \sum_{\rho} 1$, which forces $\deg \rho = 1$ for all ρ . $\square$

4.4 Isotypic Components

Theorem 4.24 (Projection onto isotypic component). *Let $V = \bigoplus_{\chi \in \text{Irr}(G)} W_\chi^{\oplus n_\chi}$ be the complete decomposition of a representation of G . For each $\chi \in \text{Irr}(G)$, the projection π_χ onto the χ -isotypic component $W_\chi^{\oplus n_\chi}$ is given by the operator*

$$\pi_\chi = \frac{\deg \chi}{\#G} \sum_{g \in G} \overline{\chi(g)} \rho(g) \in \text{End}(V).$$

Proof. Let $T = \frac{\deg \chi}{\#G} \sum_{g \in G} \overline{\chi(g)} \rho(g)$. By the lemma on morphisms from class functions, $T \in \text{End}_G(V)$, and by Schur's lemma it acts on each irreducible summand W_ψ by a scalar λ . Taking traces, $\lambda = \frac{\deg \chi}{\#G} \cdot \frac{\#G}{\deg \psi} \cdot (\psi \mid \chi) = \begin{cases} 1 & \text{if } \psi = \chi, \\ 0 & \text{if } \psi \neq \chi. \end{cases}$ So T is the identity on $W_\chi^{\oplus n_\chi}$ and zero on all other isotypic components. $\square$

Corollary 4.25 (Isotypic components are canonical). *For each χ , the isotypic component $W_\chi^{\oplus n_\chi}$ does not depend on the chosen decomposition of V . It is called the χ -isotypic component of V .*

Proof. By Theorem 4.24, the projection π_χ onto the χ -isotypic component is given by the explicit formula $\pi_\chi = \frac{\deg \chi}{\#G} \sum_{g \in G} \overline{\chi(g)} \rho(g)$, which involves only V , the representation ρ , and the character χ — not any chosen decomposition. Hence $W_\chi^{\oplus n_\chi} = \text{Im } \pi_\chi$ is canonically defined. $\square$

4.5 The Character Table

Definition 4.26 (Character table). The *character table* of a finite group G is the square matrix whose rows are indexed by $\text{Irr}(G)$, whose columns are indexed by the conjugacy classes of G , and whose (i, j) entry is the value $\chi_i(g_j)$ of the i -th irreducible character on any element g_j of the j -th conjugacy class. It is a square matrix of size $\# \text{Irr}(G) = \# \{\text{conj. classes}\}$.

Example 4.27 (Character table of S_3). From Example 3.41, with conjugacy classes represented by Id , (12) , (123) :

# elements	Id	(12)	(123)
1	1	3	2
ε	1	1	1
$\leq$	1	-1	1
	2	0	-1

Theorem 4.28 (Second orthogonality of characters). *Let C, C' be conjugacy classes of G , and let $g \in C, h \in C'$. Then*

$$\sum_{\chi \in \text{Irr}(G)} \overline{\chi(g)} \chi(h) = \begin{cases} \#G / \#C & \text{if } C = C', \\ 0 & \text{if } C \neq C'. \end{cases}$$

In particular, the columns of the character table are pairwise orthogonal.

Proof. By Theorem 4.21, $\text{Irr}(G)$ is an orthonormal basis of the space of class functions. The class indicator $\delta_C: G \rightarrow \{0, 1\}$ (equal to 1 on C and 0 elsewhere) is a class function, so it expands as

$$\delta_C = \sum_{\chi \in \text{Irr}(G)} (\delta_C | \chi) \chi.$$

The coefficient is $(\delta_C | \chi) = \frac{1}{\#G} \sum_{x \in C} \overline{\chi(x)} = \frac{\#C}{\#G} \overline{\chi(g)}$. Evaluating at h :

$$\delta_C(h) = \frac{\#C}{\#G} \sum_{\chi \in \text{Irr}(G)} \overline{\chi(g)} \chi(h).$$

Since $\delta_C(h) = 1$ if $h \in C$ (i.e. $C = C'$) and 0 otherwise, dividing by $\#C/\#G$ gives the result. $\square$

4.6 The Conjugacy Principle and Conjugacy Classes in S_n

Remark 4.29 (Conjugacy principle). In a group G of “transformations” acting on objects with “parameters” $x, y, \dots$, conjugation by $h \in G$ replaces parameters by their images under h : hgh^{-1} is the transformation of the same type as g but applied to the parameters $h \cdot x, h \cdot y, \dots$

Example 4.30. In $G = \text{SO}_3(\mathbb{R})$, if g is the rotation of axis ℓ and angle θ , then hgh^{-1} is the rotation of axis $h(\ell)$ and the same angle θ .

Applying the conjugacy principle to $G = S_n$ gives a complete description of conjugacy classes.

Proposition 4.31. Two permutations $\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type. Hence conjugacy classes of S_n are in bijection with partitions of n .

Proof. If $\sigma = (i_1 \cdots i_k)(j_1 \cdots j_l) \cdots$, then $\tau\sigma\tau^{-1} = (\tau(i_1) \cdots \tau(i_k))(\tau(j_1) \cdots \tau(j_l)) \cdots$, which has the same cycle shape. Conversely, any two permutations of the same cycle shape are conjugate via an explicit permutation of the labels. $\square$

Example 4.32. The conjugacy classes of S_5 correspond to partitions of 5:

$\{\text{Id}\}$	$5 = 1 + 1 + 1 + 1 + 1$
$\{(**)\}$	$5 = 2 + 1 + 1 + 1$
$\{(***)\}$	$5 = 3 + 1 + 1$
$\{(***)\}$	$5 = 4 + 1$
$\{(***)\}$	$5 = 5$
$\{(**)(**)\}$	$5 = 2 + 2 + 1$
$\{(***)(**)\}$	$5 = 3 + 2$

giving 7 conjugacy classes and hence 7 irreducible representations.

4.7 Example: Character Table of S_4

The conjugacy classes of S_4 correspond to partitions of 4, giving 5 classes, represented by Id , (12) , (123) , (1234) , $(12)(34)$ with class sizes 1, 6, 8, 6, 3 respectively.

There are thus 5 irreducible representations of degrees $n_1 \leq n_2 \leq n_3 \leq n_4 \leq n_5$. Since $\mathbf{1}$ is always irreducible, $n_1 = 1$. The constraint $\sum_i n_i^2 = \#S_4 = 24$ forces $n_2^2 + n_3^2 + n_4^2 + n_5^2 = 23$. If $n_5 = 4$, the remaining sum is 7, which is impossible with integers ≥ 1 . So $n_5 = 3$, leaving $n_2^2 + n_3^2 + n_4^2 = 14$. Since $3 \leq n_4 \leq 3$, we get $n_4 = 3$ and $n_2^2 + n_3^2 = 5$, so $n_2 = 1$ and $n_3 = 2$. The five degrees are therefore $(1, 1, 2, 3, 3)$.

We build the table using known representations:

- $\mathbf{1}$: trivial representation (row of all 1s).
- ε : sign representation (+1 on even permutations, -1 on odd).
- Perm: permutation representation on $\{1, 2, 3, 4\}$, with $\chi(g) = \# \text{Fix } g$. One computes $(\text{Perm} \mid \text{Perm}) = 2$ and $(\text{Perm} \mid \mathbf{1}) = 1$, so $\text{Perm} \simeq \mathbf{1} \oplus \chi$ where χ is the irreducible of degree 3 with values $(3, 1, 0, -1, -1)$.
- $\chi\varepsilon$: tensor product of χ with the sign, giving another irreducible of degree 3 with values $(3, -1, 0, 1, -1)$.
- The degree-2 representation ψ : found by subtracting. The character of $\text{Hom}(\chi, \chi)$ is $\chi\bar{\chi} = \chi^2$, with values $(9, 1, 0, 1, 1)$. Since $(\chi^2 \mid \chi^2) = 4$, $(\chi^2 \mid \mathbf{1}) = 1$, $(\chi^2 \mid \chi) = 1$, $(\chi^2 \mid \chi\varepsilon) = 1$, we get $\chi^2 \simeq \mathbf{1} \oplus \chi \oplus \chi\varepsilon \oplus \psi$ where ψ is the remaining irreducible of degree 2 with values $(2, 0, -1, 0, 2)$.

The complete character table of S_4 is:

	Id	(12)	(123)	(1234)	(12)(34)
# elements	1	6	8	6	3
$\mathbf{1}$	1	1	1	1	1
ε	1	-1	1	-1	1
ψ	2	0	-1	0	2
χ	3	1	0	-1	-1
$\chi\varepsilon$	3	-1	0	1	-1

One may verify: each row has inner product 1 with itself and 0 with all other rows; Corollary 4.4 gives $1 + 1 + 4 + 9 + 9 = 24 = \#S_4$. ✓

4.8 Representations of Abelian Groups

Since every finite abelian group is a direct product of cyclic groups, it suffices to understand the irreducible characters of $\mathbb{Z}_n$ and how characters behave under direct products. Because all irreducible representations of an abelian group have degree 1, they are simply group homomorphisms $G \rightarrow \mathbb{C}^\times$.

Definition 4.33. Let $n \in \mathbb{N}$ and $\omega_n = e^{2\pi i/n}$. For each $k = 0, 1, \dots, n-1$ define

$$\chi_k: \mathbb{Z}_n \longrightarrow \mathbb{C}^\times, \quad \chi_k(\bar{m}) = \omega_n^{km}.$$

Proposition 4.34. The characters $\chi_0, \dots, \chi_{n-1}$ are exactly the irreducible representations of $\mathbb{Z}_n$.

Proof. Each χ_k is a group homomorphism (hence an irreducible degree-1 representation). They are pairwise distinct since $\chi_k(\bar{1}) = \omega_n^k$ takes n different values. Since $\mathbb{Z}_n$ is abelian, $\# \text{Irr}(\mathbb{Z}_n) = n$, so these are all. $\square$

Example 4.35. The character table of $\mathbb{Z}_4$ (with $\omega = e^{2\pi i/4} = i$):

	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

Both orthogonality relations can be verified directly.

Proposition 4.36 (Characters of direct products). *Let G_1, G_2 be finite abelian groups with $\text{Irr}(G_1) = \{\chi_1, \dots, \chi_m\}$ and $\text{Irr}(G_2) = \{\varphi_1, \dots, \varphi_n\}$ (where $m = \#G_1, n = \#G_2$). The functions $\alpha_{ij}: G_1 \times G_2 \rightarrow \mathbb{C}^\times$ defined by*

$$\alpha_{ij}(g_1, g_2) = \chi_i(g_1) \varphi_j(g_2)$$

form a complete set of $mn = \#(G_1 \times G_2)$ irreducible representations of $G_1 \times G_2$.

Proof. Each α_{ij} is a group homomorphism, hence irreducible. They are pairwise distinct: $\alpha_{ij} = \alpha_{kl}$ implies $\chi_i = \chi_k$ (restrict to $G_1 \times \{1\}$) and $\varphi_j = \varphi_l$. Since $G_1 \times G_2$ is abelian of order mn , it has exactly mn irreducible representations. $\square$

Example 4.37. The character table of the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$, with $\text{Irr}(\mathbb{Z}_2) = \{\chi_0 \equiv 1, \chi_1(\bar{m}) = (-1)^m\}$:

	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
α_{00}	1	1	1	1
α_{01}	1	-1	1	-1
α_{10}	1	1	-1	-1
α_{11}	1	-1	-1	1

Chapter 5

Fourier Analysis on Finite Groups

Classical Fourier analysis decomposes a periodic function on $\mathbb{R}$ as a sum of pure exponentials $e^{2\pi i n x}$. The same idea applies to finite groups: every function $G \rightarrow \mathbb{C}$ expands uniquely in terms of *matrix coefficients* of irreducible representations. For abelian groups this specialises to the classical discrete Fourier transform; for non-abelian groups it yields a ring isomorphism $\mathbb{C}[G] \simeq \prod_{\rho} \mathcal{M}_{\deg \rho}(\mathbb{C})$, recovering the Wedderburn theorem from a Fourier-analytic viewpoint.

5.1 The Function Algebra $L(G)$

Definition 5.1. The *function space* $L(G) = \text{Map}(G, \mathbb{C})$ is the $\mathbb{C}$ -vector space of all functions $G \rightarrow \mathbb{C}$. It has dimension $\#G$, with basis the *delta functions* δ_g ($g \in G$), where

$$\delta_g(x) = \begin{cases} 1 & x = g, \\ 0 & \text{otherwise.} \end{cases}$$

The space $L(G)$ carries the inner product $(a \mid b) = \frac{1}{\#G} \sum_{g \in G} a(g) \overline{b(g)}$.

Definition 5.2 (Convolution). For $a, b \in L(G)$, the *convolution* $a * b \in L(G)$ is defined by

$$(a * b)(x) = \sum_{y \in G} a(xy^{-1}) b(y).$$

Proposition 5.3. For all $g, h \in G$, $\delta_g * \delta_h = \delta_{gh}$.

Proof. $(\delta_g * \delta_h)(x) = \sum_y \delta_g(xy^{-1}) \delta_h(y)$; the only non-zero term is $y = h$, which contributes $\delta_g(xh^{-1}) = 1$ iff $x = gh$. $\square$

Theorem 5.4. $(L(G), +, *)$ is a unital ring with identity δ_{1_G} , isomorphic to $\mathbb{C}[G]$ via $e_g \mapsto \delta_g$. Class functions on G are exactly the elements of the centre $Z(L(G))$.

Proof. Since $\delta_g * \delta_h = \delta_{gh}$, the assignment $e_g \mapsto \delta_g$ extends linearly to a ring isomorphism $\mathbb{C}[G] \xrightarrow{\sim} L(G)$. The characterisation of $Z(L(G))$ then follows from Proposition 4.18. $\square$

Remark 5.5. Under the isomorphism $\mathbb{C}[G] \simeq L(G)$, the element $\sum_g \lambda_g e_g$ corresponds to the function $a : g \mapsto \lambda_g$, and the group ring multiplication corresponds to convolution. We will freely switch between the two viewpoints.

5.2 Matrix Coefficients

Definition 5.6 (Matrix coefficients). Fix unitary representatives $\rho^{(1)}, \dots, \rho^{(s)}$ of all isomorphism classes in $\text{Irr}(G)$, with $d_k = \deg \rho^{(k)}$. The *matrix coefficients* of $\rho^{(k)}$ are the d_k^2 functions

$$\varphi_{ij}^{(k)} : G \longrightarrow \mathbb{C}, \quad g \longmapsto \rho^{(k)}(g)_{ij}, \quad 1 \leq i, j \leq d_k.$$

In particular, $\chi_k = \sum_i \varphi_{ii}^{(k)}$.

Theorem 5.7 (Schur orthogonality of matrix coefficients). *For all k, l and all index pairs $(a, b), (c, d)$,*

$$(\varphi_{ab}^{(k)} \mid \varphi_{cd}^{(l)}) = \frac{\delta_{kl} \delta_{ac} \delta_{bd}}{d_k}.$$

Equivalently, $\{\sqrt{d_k} \varphi_{ij}^{(k)}\}_{k,i,j}$ is an orthonormal set in $L(G)$.

Proof. For any linear map $A : V_l \rightarrow V_k$, the average

$$\tilde{A} = \frac{1}{\#G} \sum_{g \in G} \rho^{(k)}(g) \circ A \circ \rho^{(l)}(g)^{-1}$$

is a G -module morphism, so by Corollary 3.3: $\tilde{A} = 0$ if $k \neq l$, and $\tilde{A} = \frac{\text{tr } A}{d_k} \text{id}$ if $k = l$.

Apply this to $A = E_{qp}$ (the matrix with 1 in row q , column p , zero elsewhere). Since $\rho^{(l)}$ is unitary, $(\rho^{(l)}(g)^{-1})_{pj} = \rho_{jp}^{(l)}(g)$, so the (i, j) -entry of $\tilde{A}$ is

$$\frac{1}{\#G} \sum_{g \in G} \rho_{iq}^{(k)}(g) \overline{\rho_{jp}^{(l)}(g)} = \begin{cases} 0 & k \neq l, \\ \frac{\delta_{qp} \delta_{ij}}{d_k} & k = l. \end{cases}$$

Renaming $(i, q) \rightarrow (a, b)$ and $(j, p) \rightarrow (c, d)$ yields the claim. $\square$

Theorem 5.8 (Peter–Weyl theorem for finite groups). *The functions $\{\sqrt{d_k} \varphi_{ij}^{(k)}\}_{k,i,j}$ form an orthonormal basis of $L(G)$.*

Proof. Orthonormality is Theorem 5.7. By Corollary 4.4, the total count is $\sum_k d_k^2 = \#G = \dim L(G)$, so they span. $\square$

Corollary 5.9. *Restricting to class functions (i.e. to $\varphi_{11}^{(k)} + \dots + \varphi_{d_k d_k}^{(k)} = \chi_k$) recovers the orthonormal basis $\text{Irr}(G)$ of the space of class functions.*

5.3 Fourier Analysis on Abelian Groups

For the rest of this section let $G = \{g_1, \dots, g_n\}$ be a finite abelian group of order n . All irreducible representations are 1-dimensional, so $\text{Irr}(G) = \{\chi_1, \dots, \chi_n\}$ with each $\chi_i: G \rightarrow \mathbb{C}^\times$ a group homomorphism. The Peter–Weyl basis of $L(G)$ reduces to $\text{Irr}(G)$ itself.

Definition 5.10 (Fourier transform). The *Fourier transform* of $f \in L(G)$ is the function $\hat{f}: G \rightarrow \mathbb{C}$ defined by

$$\hat{f}(g_i) = n(\chi_i | f) = \sum_{g \in G} \overline{\chi_i(g)} f(g).$$

Theorem 5.11 (Fourier inversion). For any $f \in L(G)$,

$$f = \frac{1}{n} \sum_{i=1}^n \hat{f}(g_i) \chi_i.$$

Proof. Expand f in the orthonormal basis $\{\chi_i\}$: $f = \sum_i (\chi_i | f) \chi_i = \frac{1}{n} \sum_i \hat{f}(g_i) \chi_i$. $\square$

Theorem 5.12 (Convolution theorem). The Fourier transform satisfies $\widehat{a * b} = \hat{a} \cdot \hat{b}$ (pointwise product). Consequently, it is a ring isomorphism

$$(L(G), +, *) \xrightarrow{\sim} (L(G), +, \cdot),$$

or equivalently $\mathbb{C}[G] \simeq \mathbb{C}^n$ as $\mathbb{C}$ -algebras.

Proof. Setting $z = xy^{-1}$ and using that χ_i is a homomorphism:

$$\begin{aligned} \widehat{a * b}(g_i) &= \sum_{x \in G} \overline{\chi_i(x)} \sum_{y \in G} a(xy^{-1})b(y) = \sum_{y \in G} b(y) \sum_{z \in G} \overline{\chi_i(zy)} a(z) \\ &= \sum_{y \in G} \overline{\chi_i(y)} b(y) \sum_{z \in G} \overline{\chi_i(z)} a(z) = \hat{b}(g_i) \hat{a}(g_i). \end{aligned}$$

The Fourier transform is a vector space isomorphism by Fourier inversion, and it sends $*$ to $\cdot$, so it is a ring isomorphism. The target ring $(\mathbb{C}^n, \cdot)$ is the product ring $\mathbb{C} \times \dots \times \mathbb{C}$, recovering the abelian case of the Wedderburn decomposition. $\square$

Example 5.13 (Discrete Fourier transform on $\mathbb{Z}_n$). Take $G = \mathbb{Z}_n$ with $\omega = e^{2\pi i/n}$ and $\chi_k(\bar{m}) = \omega^{km}$ (Proposition 4.34). For $f: \mathbb{Z}_n \rightarrow \mathbb{C}$,

$$\hat{f}(\bar{k}) = \sum_{m=0}^{n-1} \omega^{-km} f(\bar{m}), \quad f(\bar{m}) = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{km} \hat{f}(\bar{k}).$$

These are the classical *discrete Fourier transform* (DFT) and its inverse. The convolution $(a * b)(\bar{m}) = \sum_{j=0}^{n-1} a(\bar{m} - \bar{j}) b(\bar{j})$ satisfies $\widehat{a * b} = \hat{a} \cdot \hat{b}$. The fast Fourier transform (FFT) computes $\hat{f}$ in $O(n \log n)$ operations by exploiting this algebraic structure.

5.4 Cayley Graphs and Eigenvalues

Definition 5.14 (Cayley graph). Let G be a finite group and $S \subseteq G$ a symmetric subset: $1_G \notin S$ and $s \in S \Rightarrow s^{-1} \in S$. The Cayley graph $\Gamma(G, S)$ has vertex set G and an edge $\{g, h\}$ whenever $gh^{-1} \in S$.

Remark 5.15. The Cayley graph is $\#S$ -regular. It is connected if and only if S generates G .

Example 5.16. Take $G = \mathbb{Z}_4$ and $S = \{\bar{1}, \bar{3}\}$. Then $\Gamma(\mathbb{Z}_4, S)$ is the 4-cycle with adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Theorem 5.17. Let $G = \{g_1, \dots, g_n\}$ be a finite abelian group, $S \subseteq G$ symmetric, and A the adjacency matrix of $\Gamma(G, S)$ (vertices in the order $g_1, \dots, g_n$). Then:

1. The eigenvalues of A are the real numbers $\lambda_i = \sum_{s \in S} \chi_i(s)$, $1 \leq i \leq n$.
2. An orthonormal basis of eigenvectors is $v_i = \frac{1}{\sqrt{n}}(\chi_i(g_1), \dots, \chi_i(g_n))^T$.

Proof. Let $F: L(G) \rightarrow L(G)$ be convolution by $\delta_S = \sum_{s \in S} \delta_s$. By the convolution theorem, χ_j is an eigenvector of F with eigenvalue $\hat{\delta}_S(g_j) = \sum_{s \in S} \overline{\chi_j(s)}$. Since $S = S^{-1}$ and $\chi_j(s^{-1}) = \overline{\chi_j(s)}$, we get $\hat{\delta}_S(g_j) = \sum_{s \in S} \chi_j(s) = \lambda_j \in \mathbb{R}$.

The matrix of F in the δ_{g_i} -basis satisfies $[F]_{ij} = (\delta_S * \delta_{g_j})(g_i) = \sum_{s \in S} \delta_{sg_j}(g_i) = \mathbf{1}[g_i g_j^{-1} \in S] = A_{ij}$, so $[F] = A$. The corresponding eigenvectors of A are $v_j = \frac{1}{\sqrt{n}}(\chi_j(g_1), \dots, \chi_j(g_n))^T$, orthonormal by the Peter–Weyl theorem. $\square$

Corollary 5.18 (Circulant matrices). If A is the adjacency matrix of $\Gamma(\mathbb{Z}_n, S)$, then with $\omega = e^{2\pi i/n}$ its eigenvalues are $\lambda_k = \sum_{s \in S} \omega^{ks}$ for $k = 0, \dots, n-1$, with eigenvector $v_k = \frac{1}{\sqrt{n}}(1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})^T$.

Example 5.19. Take $G = \mathbb{Z}_6$ and $S = \{\pm\bar{1}, \pm\bar{2}\}$ with $\omega = e^{2\pi i/6}$. The eigenvalues of the adjacency matrix are

$$\lambda_k = 2 \cos \frac{\pi k}{3} + 2 \cos \frac{2\pi k}{3}, \quad k = 0, 1, \dots, 5.$$

5.5 Fourier Analysis on Non-Abelian Groups

For a non-abelian group G , the irreducible representations have degrees $d_k \geq 1$ and the Fourier transform takes values in a product of matrix algebras.

Definition 5.20 (Non-abelian Fourier transform). Let $\rho^{(1)}, \dots, \rho^{(s)}$ be unitary representatives of

$\text{Irr}(G)$ with $d_k = \deg \rho^{(k)}$, and let $n = \#G$. The *Fourier transform* of $f \in L(G)$ is the s -tuple

$$T(f) = (\hat{f}(\rho^{(1)}), \dots, \hat{f}(\rho^{(s)})) \in \prod_{k=1}^s \mathcal{M}_{d_k}(\mathbb{C}),$$

where $\hat{f}(\rho^{(k)})$ is the $d_k \times d_k$ matrix with (i, j) -entry $\sum_{g \in G} \overline{\rho_{ij}^{(k)}(g)} f(g)$.

Remark 5.21. For abelian G , all $d_k = 1$ and $\hat{f}(\rho^{(k)}) = \hat{f}(g_k) \in \mathbb{C}$, recovering Definition 5.10.

Remark 5.22 (Fourier transform of a delta function). For the delta function δ_g , the definition gives

$$\hat{\delta}_g(\rho^{(k)})_{ij} = \sum_{x \in G} \overline{\rho_{ij}^{(k)}(x)} \delta_g(x) = \overline{\rho_{ij}^{(k)}(g)},$$

so $\hat{\delta}_g(\rho^{(k)}) = \overline{\rho^{(k)}(g)}$ (entry-wise conjugate of the matrix $\rho^{(k)}(g)$). Thus

$$T(\delta_g) = (\overline{\rho^{(1)}(g)}, \dots, \overline{\rho^{(s)}(g)}).$$

The tuple $T(\delta_g)$ packages the images of g under every irreducible representation, encoding the same data as the row of the character table at g : taking traces gives $\text{tr}(\hat{\delta}_g(\rho^{(k)})) = \overline{\chi_k(g)}$.

Theorem 5.23 (Fourier inversion). For any $f \in L(G)$ and $g \in G$,

$$f(g) = \frac{1}{n} \sum_{k=1}^s d_k \sum_{i,j} \hat{f}(\rho^{(k)})_{ij} \varphi_{ij}^{(k)}(g).$$

Proof. Expand f in the Peter–Weyl basis (Theorem 5.8):

$$f = \sum_{k,i,j} (f \mid \sqrt{d_k} \varphi_{ij}^{(k)}) \sqrt{d_k} \varphi_{ij}^{(k)} = \sum_{k,i,j} d_k (f \mid \varphi_{ij}^{(k)}) \varphi_{ij}^{(k)}.$$

The coefficient is $(f \mid \varphi_{ij}^{(k)}) = \frac{1}{n} \sum_g f(g) \overline{\rho_{ij}^{(k)}(g)} = \frac{1}{n} \hat{f}(\rho^{(k)})_{ij}$. Evaluating at g gives the result. $\square$

Example 5.24 (Fourier transform on S_3). Let $G = S_3$, $n = 6$. There are three irreducible unitary representations:

- $\rho^{(1)}$: trivial representation, $d_1 = 1$;
- $\rho^{(2)}$: sign representation, $d_2 = 1$, with $\rho^{(2)}(\sigma) = \text{sgn}(\sigma)$;
- $\rho^{(3)}$: standard representation, $d_3 = 2$, acting on the subspace $V = \{x_1 + x_2 + x_3 = 0\} \subset \mathbb{C}^3$ by permuting coordinates. In the orthonormal basis $u_1 = \frac{1}{\sqrt{2}}(e_1 - e_2)$, $u_2 = \frac{1}{\sqrt{6}}(e_1 + e_2 - 2e_3)$ the matrices at three representative elements are

$$\rho^{(3)}(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho^{(3)}((12)) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho^{(3)}((123)) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

Take $f = \delta_{(12)} \in L(S_3)$. Since all entries of $\rho^{(3)}$ are real, Remark 5.22 gives

$$T(f) = \left(\underbrace{1}_{\hat{f}(\rho^{(1)})}, \underbrace{-1}_{\hat{f}(\rho^{(2)})}, \underbrace{\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}}_{\hat{f}(\rho^{(3)})} \right).$$

The degree-2 component is a non-scalar matrix — a feature impossible for abelian groups.

We verify Fourier inversion (Theorem 5.23) at two points. At $g = (12)$ (using only the diagonal entries of $\hat{f}(\rho^{(3)})$, since off-diagonal entries are zero):

$$\frac{1}{6} [1 \cdot (1)(1) + 1 \cdot (-1)(-1) + 2((-1)(-1) + (1)(1))] = \frac{1}{6}(1 + 1 + 4) = 1 = f((12)). \checkmark$$

At $g = (123)$ (where $\text{sgn}((123)) = +1$ and $\rho_{11}^{(3)}((123)) = \rho_{22}^{(3)}((123)) = -\frac{1}{2}$):

$$\frac{1}{6} [1 \cdot 1 + (-1) \cdot 1 + 2((-1)(-\frac{1}{2}) + (1)(-\frac{1}{2}))] = \frac{1}{6}(1 - 1 + 0) = 0 = f((123)). \checkmark$$

Theorem 5.25 (Wedderburn via Fourier). *The Fourier transform $T: L(G) \rightarrow \prod_{k=1}^s \mathcal{M}_{d_k}(\mathbb{C})$ is a ring isomorphism (with pointwise matrix multiplication on the right).*

Proof. T is a vector space isomorphism by Fourier inversion (it is injective and both spaces have dimension $\sum_k d_k^2 = n$). It remains to check $T(a * b) = T(a) \cdot T(b)$, i.e., $\widehat{a * b}(\rho^{(k)}) = \hat{a}(\rho^{(k)})\hat{b}(\rho^{(k)})$ for each k . Setting $z = xy^{-1}$ and using that $\rho^{(k)}$ is a homomorphism:

$$\begin{aligned} \widehat{a * b}(\rho^{(k)})_{ij} &= \sum_x \overline{\rho_{ij}^{(k)}(x)} \sum_y a(xy^{-1})b(y) = \sum_y b(y) \sum_z \overline{\rho_{ij}^{(k)}(zy)} a(z) \\ &= \sum_y b(y) \sum_z \sum_m \overline{\rho_{im}^{(k)}(z)} \rho_{mj}^{(k)}(y) a(z) = \sum_m \hat{a}(\rho^{(k)})_{im} \hat{b}(\rho^{(k)})_{mj}. \end{aligned}$$

This is the (i, j) -entry of $\hat{a}(\rho^{(k)})\hat{b}(\rho^{(k)})$, as required. $\square$

Remark 5.26. This is the Wedderburn decomposition $\mathbb{C}[G] \simeq \prod_{\rho \in \text{Irr } G} \mathcal{M}_{\deg \rho}(\mathbb{C})$ from Theorem 4.15, now proved via Fourier analysis rather than module theory. The Fourier inversion formula also recovers the isotypic projections of Theorem 4.24: the element $\pi_\rho = \frac{\deg \rho}{\#G} \sum_g \chi_\rho(g) e_g$ in $\mathbb{C}[G]$ corresponds to the tuple with the identity matrix in the ρ -slot and zero elsewhere.

Chapter 6

Burnside's Theorem

In this chapter we prove one of the first major applications of representation theory: Burnside's pq -theorem, which states that a group of order $p^a q^b$ (with p, q prime) is never simple unless it is cyclic of prime order. The proof requires a brief excursion into algebraic number theory, specifically the arithmetic of algebraic integers and a touch of Galois theory of cyclotomic fields.

6.1 A Little Number Theory

Definition 6.1 (Algebraic integer). A complex number α is called an *algebraic integer* if it is a root of a monic polynomial with integer coefficients: there exists $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ with $a_0, \dots, a_{n-1} \in \mathbb{Z}$ and $p(\alpha) = 0$.

Example 6.2 (n th roots). Let $m \in \mathbb{Z}$. The polynomial $z^n - m$ is monic with integer coefficients, so every n th root of m is an algebraic integer. In particular $\sqrt{2}$ is an algebraic integer, as is $e^{2\pi i/n}$; in fact every n th root of unity is an algebraic integer.

Example 6.3 (Eigenvalues of integer matrices). Let $A = (a_{ij})$ be an $n \times n$ matrix with integer entries $a_{ij} \in \mathbb{Z}$. The characteristic polynomial $p_A(z) = \det(zI - A)$ is monic with integer coefficients, so every eigenvalue of A is an algebraic integer.

A rational number such as $1/2$ satisfies $2z - 1 = 0$, a polynomial with integer coefficients but non-unit leading coefficient, yet $1/2$ is *not* an algebraic integer. The requirement that the leading coefficient be 1 is essential.

Proposition 6.4. A rational number r is an algebraic integer if and only if $r \in \mathbb{Z}$.

Proof. Every integer r satisfies $z - r = 0$, so integers are algebraic integers. Conversely, write $r = m/n$ with $m, n \in \mathbb{Z}$, $n > 0$, and $\gcd(m, n) = 1$. Suppose r satisfies $z^k + a_{k-1}z^{k-1} + \cdots + a_0 = 0$ with $a_i \in \mathbb{Z}$. Multiplying through by n^k :

$$m^k + a_{k-1}m^{k-1}n + \cdots + a_1mn^{k-1} + a_0n^k = 0.$$

Hence $m^k = -n(a_{k-1}m^{k-1} + \cdots + a_0n^{k-1})$, so $n \mid m^k$. Since $\gcd(m, n) = 1$, we conclude $n = 1$ and $r = m \in \mathbb{Z}$. $\square$

A useful strategy for showing that an integer d divides an integer n is to show that n/d is an algebraic integer; Proposition 6.4 then forces $n/d \in \mathbb{Z}$. For this to work, we need algebraic integers to be closed under arithmetic operations, which we prove via the following characterisation.

Lemma 6.5. *An element $y \in \mathbb{C}$ is an algebraic integer if and only if there exist $y_1, \dots, y_t \in \mathbb{C}$, not all zero, such that*

$$y y_i = \sum_{j=1}^t a_{ij} y_j \quad \text{for all } 1 \leq i \leq t,$$

with $a_{ij} \in \mathbb{Z}$ (i.e. $y y_i$ is an integral linear combination of $y_1, \dots, y_t$ for each i).

Proof. Suppose y is an algebraic integer, a root of $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$. Take $y_i = y^{i-1}$ for $1 \leq i \leq n$. Then $y y_i = y^i = y_{i+1}$ for $1 \leq i \leq n-1$, and $y y_n = y^n = -a_0 y_1 - a_1 y_2 - \dots - a_{n-1} y_n$ (since y satisfies p). Since $y_1 = 1 \neq 0$, the elements are not all zero.

Conversely, suppose such $y_1, \dots, y_t$ exist. Let $A = (a_{ij})$ and $Y = (y_1, \dots, y_t)^\top \in \mathbb{C}^t$. The condition reads $[AY]_i = \sum_j a_{ij} y_j = y y_i = y[Y]_i$, so $AY = yY$. Since $Y \neq 0$, y is an eigenvalue of the $t \times t$ integer matrix A , hence an algebraic integer by Example 6.3. $\square$

Corollary 6.6. *The set $\mathbb{A}$ of algebraic integers is a subring of $\mathbb{C}$. In particular, sums and products of algebraic integers are algebraic integers. Moreover, the complex conjugate of an algebraic integer is an algebraic integer.*

Proof. If $p(z)$ is monic with integer coefficients and $p(\alpha) = 0$, then $(-1)^n p(-z)$ is a monic polynomial with integer coefficients vanishing at $-\alpha$, so $-\alpha \in \mathbb{A}$.

Let $y, y' \in \mathbb{A}$ and choose witnesses $y_1, \dots, y_t$ and $y'_1, \dots, y'_s$ as in Lemma 6.5. The products $\{y_i y'_k : 1 \leq i \leq t, 1 \leq k \leq s\}$ are not all zero. For the sum:

$$(y + y') y_i y'_k = y y_i \cdot y'_k + y' y'_k \cdot y_i = \sum_j a_{ij} y_j y'_k + \sum_\ell b_{k\ell} y'_\ell y_i,$$

an integral linear combination of the products $y_j y'_\ell$, so $y + y' \in \mathbb{A}$ by Lemma 6.5. For the product:

$$y y' \cdot y_i y'_k = (y y_i)(y' y'_k) = \left(\sum_j a_{ij} y_j \right) \left(\sum_\ell b_{k\ell} y'_\ell \right) = \sum_{j,\ell} a_{ij} b_{k\ell} y_j y'_\ell,$$

so $y y' \in \mathbb{A}$. For conjugation: if $p(\alpha) = 0$ with $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$ and $a_i \in \mathbb{Z} \subseteq \mathbb{R}$, then $p(\bar{\alpha}) = \overline{p(\alpha)} = 0$. $\square$

6.2 The Dimension Theorem

The connection between algebraic integers and representation theory begins with the following observation.

Corollary 6.7. *Let χ be a character of a finite group G . Then $\chi(g)$ is an algebraic integer for all $g \in G$.*

Proof. Let $\varphi: G \rightarrow \text{GL}_m(\mathbb{C})$ be a representation with character χ . By Lemma 3.22, φ_g is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_m$ that are n th roots of unity (where $n = \#G$). Roots of unity are algebraic integers (Example 6.2), and $\mathbb{A}$ is closed under addition (Corollary 6.6), so $\chi(g) = \lambda_1 + \dots + \lambda_m \in \mathbb{A}$. $\square$

Remark 6.8. The proof shows that $\chi_\varphi(g)$ is a sum of exactly $\deg \varphi$ roots of unity of order dividing $\#G$. We shall use this precision later.

The key algebraic fact needed for the dimension theorem is:

Theorem 6.9. *Let φ be an irreducible representation of G of degree d , let $g \in G$, and let $h = \#C$ where C is the conjugacy class of g . Then $(h/d) \chi_\varphi(g)$ is an algebraic integer.*

Proof. Let $C_1, \dots, C_s$ be the conjugacy classes of G with sizes $h_i = \#C_i$, and let χ_i denote the value of χ_φ on C_i . For each i define the *class sum*

$$T_i = \sum_{x \in C_i} \varphi_x \in \text{End}(V).$$

We establish two claims.

Claim 1. $T_i = \frac{h_i \chi_i}{d} I$.

For any $g \in G$, $\varphi_g T_i \varphi_g^{-1} = \sum_{x \in C_i} \varphi_{gxg^{-1}} = \sum_{y \in C_i} \varphi_y = T_i$ (since C_i is stable under conjugation). Thus T_i commutes with all φ_g , so by Corollary 3.3 we have $T_i = \lambda I$ for some $\lambda \in \mathbb{C}$. Taking traces: $d\lambda = \text{tr}(T_i) = \sum_{x \in C_i} \chi_i = h_i \chi_i$, giving $\lambda = h_i \chi_i / d$.

Claim 2. $T_i T_j = \sum_{k=1}^s a_{ijk} T_k$ for non-negative integers a_{ijk} .

Direct computation gives $T_i T_j = \sum_{g \in G} a_{ijg} \varphi_g$ where $a_{ijg} = \#\{(x, y) \in C_i \times C_j : xy = g\}$. If $g' = kgk^{-1}$, the map $(x, y) \mapsto (kxk^{-1}, kyk^{-1})$ is a bijection from $\{xy = g\}$ to $\{xy = g'\}$ inside $C_i \times C_j$, so a_{ijg} depends only on the conjugacy class of g . Writing a_{ijk} for the common value on C_k gives $T_i T_j = \sum_k a_{ijk} T_k$.

Conclusion. Set $\beta_k = h_k \chi_k / d$ for $k = 1, \dots, s$. Substituting Claim 1 into Claim 2 gives $\beta_i \beta_k = \sum_j a_{ijk} \beta_j$ for each i, k . Fixing i , the elements $\beta_1, \dots, \beta_s$ satisfy $\beta_i \cdot \beta_k = \sum_j a_{ijk} \beta_j$, an integral linear combination of $\beta_1, \dots, \beta_s$. Since $\beta_1 = h_1 \chi_1 / d = 1 \cdot d / d = 1 \neq 0$, the β_j are not all zero. By Lemma 6.5, each $\beta_i = h_i \chi_i / d$ is an algebraic integer. $\square$

Theorem 6.10 (Dimension theorem). *Let φ be an irreducible representation of a finite group G of degree d . Then d divides $\#G$.*

Proof. Write $n = \#G$. By the first orthogonality relations (Theorem 3.35), $\langle \chi_\varphi, \chi_\varphi \rangle = 1$, i.e.

$$\frac{n}{d} = \sum_{g \in G} \overline{\chi_\varphi(g)} \cdot \frac{\chi_\varphi(g)}{d}.$$

Grouping by conjugacy classes $C_1, \dots, C_s$ with sizes h_i and character values χ_i :

$$\frac{n}{d} = \sum_{i=1}^s \overline{\chi_i} \cdot \frac{h_i \chi_i}{d}.$$

Each $h_i \chi_i / d$ is an algebraic integer by Theorem 6.9; each $\overline{\chi_i}$ is an algebraic integer by Corollary 6.7 and Corollary 6.6. Since $\mathbb{A}$ is a ring, each summand is algebraic, and so is the sum n/d . Being rational, Proposition 6.4 gives $n/d \in \mathbb{Z}$, i.e. $d \mid n$. $\square$

Corollary 6.11. *If $\#G = p^2$ for a prime p , then G is abelian.*

Proof. Each irreducible degree d_i divides p^2 , so $d_i \in \{1, p, p^2\}$. If $d_i \geq p$ for some i , then $d_i^2 \geq p^2 = \#G$, leaving $\sum_{j \neq i} d_j^2 = 0$ (impossible since the trivial representation contributes $d_j = 1$). Hence all $d_i = 1$. Then $\# \text{Irr}(G) = \sum_i 1 = \sum_i d_i^2 = p^2 = \#G$, so by Theorem 4.21 there are p^2 conjugacy classes and p^2 elements; every conjugacy class is a singleton, meaning every element is central. Thus $G = Z(G)$ is abelian. $\square$

Recall that the *commutator subgroup* G' of G is the subgroup generated by all commutators $g^{-1}h^{-1}gh$ with $g, h \in G$. It is normal, G/G' is abelian, and if $N \trianglelefteq G$ with G/N abelian then $G' \subseteq N$.

Lemma 6.12. *The number of degree-one irreducible representations of G equals $[G : G']$, and this number divides $\#G$.*

Proof. A degree-one representation $\rho: G \rightarrow \mathbb{C}^*$ has abelian image, so $G' \subseteq \ker \rho$ and ρ factors as $\rho = \psi \circ \pi$ where $\pi: G \rightarrow G/G'$ is the quotient map and ψ is a degree-one representation of G/G' . Conversely, every degree-one representation of G/G' pulls back to one of G this way. Since G/G' is abelian, every irreducible representation of G/G' is one-dimensional (in an irreducible representation of an abelian group, every element acts as a scalar by Schur's lemma Corollary 3.3, so every one-dimensional subspace is invariant and the space must be one-dimensional). By Theorem 4.21 applied to G/G' , there are $\#(G/G') = [G : G']$ such representations, which divides $\#G$. $\square$

Corollary 6.13. *Let $p < q$ be primes with $q \not\equiv 1 \pmod{p}$. Then every group of order pq is abelian (hence cyclic).*

Proof. Let $d_1, \dots, d_s$ be the irreducible degrees. Each d_i divides pq and satisfies $d_i^2 \leq pq$; since $p < q$, both $q^2 > pq$ and $(pq)^2 > pq$, so $d_i \in \{1, p\}$. Let m be the number of degree-1 representations and r the number of degree- p representations. Then $m + p^2r = pq$, so $p \mid m$. By Lemma 6.12, $m \mid pq$; combined with $p \mid m$, we get $m \in \{p, pq\}$. If $m = p$, then $p^2r = p(q - 1)$, giving $pr = q - 1$, i.e. $p \mid q - 1$, meaning $q \equiv 1 \pmod{p}$, contrary to hypothesis. Hence $m = pq$ and G is abelian. $\square$

6.3 Burnside's Theorem

The proof of Burnside's theorem uses Galois theory of cyclotomic fields. We develop what we need from scratch.

Recall from Remark 6.8 that $\chi_\varphi(g)$ is a sum of $d = \deg \varphi$ roots of unity of order dividing $n = \#G$. The following elementary lemma controls the absolute value of such sums.

Lemma 6.14. *Let $\lambda_1, \dots, \lambda_d$ be roots of unity. Then $|\lambda_1 + \dots + \lambda_d| \leq d$, with equality if and only if $\lambda_1 = \dots = \lambda_d$.*

Proof. The triangle inequality gives $|\lambda_1 + \dots + \lambda_d| \leq |\lambda_1| + \dots + |\lambda_d| = d$. Equality holds if and only if all non-zero summands point in the same direction, i.e. all λ_i are non-negative real multiples of the same complex number. Since $|\lambda_i| = 1$, this forces $\lambda_1 = \dots = \lambda_d$. $\square$

Let $\omega = e^{2\pi i/n}$ and let $\Gamma = \text{Gal}(\mathbb{Q}[\omega]/\mathbb{Q})$ be the group of all field automorphisms $\sigma: \mathbb{Q}[\omega] \rightarrow \mathbb{Q}[\omega]$ that fix $\mathbb{Q}$ pointwise. By the fundamental theorem of Galois theory, $|\Gamma| = \phi(n)$ (Euler's totient function) and $\Gamma \cong \mathbb{Z}_n^*$.

Lemma 6.15. *Let $p(z)$ be a polynomial with rational coefficients and let $\alpha \in \mathbb{Q}[\omega]$ be a root of p . Then $\sigma(\alpha)$ is also a root of p for every $\sigma \in \Gamma$.*

Proof. Write $p(z) = \sum a_k z^k$ with $a_k \in \mathbb{Q}$. Then $p(\sigma(\alpha)) = \sum a_k \sigma(\alpha)^k = \sigma(\sum a_k \alpha^k) = \sigma(p(\alpha)) = \sigma(0) = 0$, using $\sigma(a_k) = a_k$ since $a_k \in \mathbb{Q}$. $\square$

Corollary 6.16. *If α is an n th root of unity then $\sigma(\alpha)$ is an n th root of unity for every $\sigma \in \Gamma$.*

Proof. Apply Lemma 6.15 to $p(z) = z^n - 1$. $\square$

Corollary 6.17. *If $\alpha \in \mathbb{Q}[\omega]$ is an algebraic integer and $\sigma \in \Gamma$, then $\sigma(\alpha)$ is an algebraic integer.*

Proof. α is a root of a monic polynomial p with integer coefficients; $\sigma(\alpha)$ is also a root by Lemma 6.15. $\square$

We use the following two standard results from Galois theory without proof.

Theorem 6.18. *Let $\alpha \in \mathbb{Q}[\omega]$. Then $\sigma(\alpha) = \alpha$ for all $\sigma \in \Gamma$ if and only if $\alpha \in \mathbb{Q}$.*

Corollary 6.19. *For any $\alpha \in \mathbb{Q}[\omega]$, the product $\prod_{\sigma \in \Gamma} \sigma(\alpha)$ lies in $\mathbb{Q}$.*

Proof. For any $\tau \in \Gamma$, applying τ permutes the factors: $\tau(\prod_{\sigma} \sigma(\alpha)) = \prod_{\sigma} \tau\sigma(\alpha) = \prod_{\rho} \rho(\alpha)$. Hence the product is fixed by all $\tau \in \Gamma$ and belongs to $\mathbb{Q}$ by Theorem 6.18. $\square$

The following theorem is the key technical engine of Burnside's argument.

Theorem 6.20. *Let G have order n , let C be a conjugacy class of G , and let $\varphi: G \rightarrow \text{GL}_d(\mathbb{C})$ be an irreducible representation with $\gcd(\#C, d) = 1$. Then for every $g \in C$, either $\varphi_g = \lambda I$ for some $\lambda \in \mathbb{C}^*$, or $\chi_\varphi(g) = 0$.*

Proof. Set $\chi = \chi_\varphi$ and $h = \#C$. Since $\gcd(h, d) = 1$, choose integers k, j with $kh + jd = 1$. Define

$$\alpha = k \cdot \frac{h}{d} \chi(g) + j \chi(g) = \frac{\chi(g)}{d}.$$

This is an algebraic integer: $(h/d)\chi(g)$ is algebraic by Theorem 6.9, $\chi(g)$ is algebraic by Corollary 6.7, and $\mathbb{A}$ is a ring (Corollary 6.6).

Since χ is a class function, it is either zero on all of C or on none. It therefore suffices to show: if φ_g is not a scalar matrix, then $\chi(g) = 0$.

Assume $\varphi_g \neq \lambda I$ for all λ . By Lemma 3.22, φ_g is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_d$ that are n th roots of unity, hence in $\mathbb{Q}[\omega]$. Since φ_g is not scalar, the λ_i are not all equal, so Lemma 6.14 gives $|\chi(g)| < d$, hence $|\alpha| < 1$.

For any $\sigma \in \Gamma$: $\sigma(\alpha) = \sigma(\chi(g))/d = (\sigma(\lambda_1) + \dots + \sigma(\lambda_d))/d$. By Corollary 6.16, each $\sigma(\lambda_i)$ is an n th root of unity. Since σ is injective and the λ_i are not all equal, neither are the $\sigma(\lambda_i)$; another application of Lemma 6.14 gives $|\sigma(\alpha)| < 1$. By Corollary 6.17, $\sigma(\alpha)$ is an algebraic integer.

Therefore $q = \prod_{\sigma \in \Gamma} \sigma(\alpha)$ is a product of algebraic integers each of modulus < 1 , so q is an algebraic integer with $|q| < 1$. By Corollary 6.19, $q \in \mathbb{Q}$, and by Proposition 6.4, $q \in \mathbb{Z}$. Since $|q| < 1$, we get $q = 0$, hence $\sigma(\alpha) = 0$ for some σ . Since σ is a field automorphism, $\alpha = 0$, i.e. $\chi(g) = 0$. $\square$

Lemma 6.21. *Let G be a finite non-abelian group. If G has a conjugacy class $C \neq \{1\}$ with $\#C$ a power of a prime p , then G is not simple.*

Proof. Suppose for contradiction that G is simple. Let $\varphi^{(1)}, \dots, \varphi^{(s)}$ be the irreducible representations of G with $\varphi^{(1)}$ trivial, characters $\chi_1, \dots, \chi_s$, and degrees $d_1 = 1, d_2, \dots, d_s$. For $k \geq 2$, the kernel $\ker \varphi^{(k)}$ is normal in G . By simplicity it is $\{1\}$ or G ; since $\varphi^{(k)}$ is non-trivial, $\ker \varphi^{(k)} = \{1\}$, so $\varphi^{(k)}$ is injective. Since G is non-abelian and $\mathbb{C}^*$ is abelian, $d_k > 1$ for all $k \geq 2$. Since G is simple and non-abelian, $Z(G) = \{1\}$, so $\#C = p^t$ with $t \geq 1$.

Let $g \in C$ and $k \geq 2$. If $p \nmid d_k$, then $\gcd(\#C, d_k) = \gcd(p^t, d_k) = 1$, so Theorem 6.20 applies: either $\varphi_g^{(k)} = \lambda I$ or $\chi_k(g) = 0$. In the former case, the set $Z_k = \{x \in G : \varphi_x^{(k)} \text{ is scalar}\}$ is the preimage of the centre of $\text{GL}_{d_k}(\mathbb{C})$ under $\varphi^{(k)}$, a normal subgroup of G . Since $d_k > 1$, $Z_k \neq G$, so $Z_k = \{1\}$ by simplicity. But $g \in C$ and $\varphi_g^{(k)}$ is scalar means $g \in Z_k = \{1\}$, contradicting $g \in C \neq \{1\}$. Therefore $\chi_k(g) = 0$ whenever $p \nmid d_k$.

The character of the regular representation (Theorem 4.3) satisfies $\chi_L(g) = 0$ for $g \neq 1$:

$$0 = 1 + \sum_{k=2}^s d_k \chi_k(g) = 1 + \sum_{\substack{k \geq 2 \\ p \mid d_k}} d_k \chi_k(g).$$

Dividing by p :

$$\sum_{\substack{k \geq 2 \\ p \mid d_k}} \frac{d_k}{p} \chi_k(g) = -\frac{1}{p}.$$

The left side is a sum of integer multiples of algebraic integers, hence an algebraic integer. Being rational, Proposition 6.4 forces it to be an integer. But $-1/p \notin \mathbb{Z}$ — a contradiction. $\square$

Theorem 6.22 (Burnside's pq -theorem). *Let G be a group of order $p^a q^b$ with p, q prime. Then G is not simple unless it is cyclic of prime order.*

Proof. An abelian group is simple if and only if it is cyclic of prime order. So we may assume G is non-abelian.

If $a = 0$ (or $b = 0$), then G is a p -group (or q -group). A non-trivial p -group has non-trivial centre $Z(G)$, which is a proper non-trivial normal subgroup when G is non-abelian, so G is not simple.

Suppose $a, b \geq 1$. By Sylow's theorem, G has a Sylow q -subgroup H of order q^b . Choose $1 \neq g \in Z(H)$ (possible since $Z(H) \neq \{1\}$ as H is a non-trivial q -group). Since every element of H commutes with g , we have $H \subseteq C_G(g) \subseteq N_G(g)$, so

$$\#C_g = [G : N_G(g)] \mid [G : H] = p^a.$$

Thus $\#C_g = p^t$ for some $0 \leq t \leq a$. Since $g \neq 1$, we have $C_g \neq \{1\}$. By Lemma 6.21, G is not simple. $\square$

Remark 6.23. Burnside's theorem is often stated in the equivalent form: every group of order $p^a q^b$ is solvable. This follows by induction: a non-simple group has a proper normal subgroup N , and both N and G/N have orders of the form $p^{a'} q^{b'}$ with $a' \leq a$, $b' \leq b$, and $\#N < \#G$, so they are solvable by induction, and hence so is G .